A Reconstruction Methodology for Marginal Cost and Marginal CO_2 Emissions in the Chilean Power System Using CEN Public Data

Marcelo Matus A., *Member, IEEE,* María del Pilar Buitrago Villada, Erick Sierra Baeza, Carlos Enrique García Bujanda, Marcos Falcone Salazar, and Gonzalo Bustos Turu

***Abstract*—The decarbonisation of the Chilean power sector and the country's commitment to a carbon-tax escalation under Law 20.780 require operational signals that are both *spatially* resolved (per-bus) and *marginally* attributed (per-unit) — the natural analogue of a Locational Marginal Emission factor. At present the Coordinador Eléctrico Nacional (CEN) publishes only a 15-minute nodal marginal cost λ^{CEN} and an annual system average emission factor; no per-bus, per-quarter marginal CO_2 intensity ε is released. This paper presents `cen2gtopt`, a novel reconstruction methodology that re-engineers ε for the 29 CEN-published 220/500 kV buses at 15-minute resolution from public data alone. The procedure combines the SCVIC merit-ladder of declared variable costs, the "generación real" hourly dispatch, the "instrucciones operacionales" price-taker classification, and the Marginal Loss Factors (MLFs) extracted from the PLEXOS PCP/PID solution `.accdb` oracle. A chained-tolerance plus cumulative-spread clustering rule separates the 29 buses into electrically coherent islands per quarter; for each island the marginal unit is identified as the dispatched non-forced generator whose declared CV is closest to the median bus marginal price. The methodology is applied to the operating day 2026-04-22. Across 2494 (bus, quarter) cells, the reconstructed marginal cost $\hat{\lambda}$ agrees with λ^{CEN} to a mean absolute error of $0.170\,\mathrm{USD\,MWh^{-1}}$, a 90$^{\mathrm{th}}$ percentile of $0.492\,\mathrm{USD\,MWh^{-1}}$ and a maximum of $2.533\,\mathrm{USD\,MWh^{-1}}$; after excluding the 13.7 % of cells flagged as `tier_uncertainty`, the mean drops to $0.137\,\mathrm{USD\,MWh^{-1}}$ and the maximum to $1.34\,\mathrm{USD\,MWh^{-1}}$. Per-island marginal-unit coherence is exact (0 of 474 islands disagree). The pipeline runs in approximately 6 s on a warm cache and is published as part of the `gtopt` project.**

***Index Terms*—Locational marginal emissions, marginal CO_2 factor, locational marginal price, Coordinador Eléctrico Nacional, SCVIC, marginal loss factor, merit ladder, public-data reconstruction, short-run-marginal emissions, Chilean power system.**

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M. Matus A. is with Black Bear Engineering SpA, Santiago, Chile, and the Centro Nacional de Inteligencia Artificial (CENIA, FB210017), Santiago, Chile (e-mail: marcelo.matus@blackbear.cl).

M. P. Buitrago Villada is with the Department of Electrical Energy and Automation, Universidad Nacional de Colombia, Medellín, Colombia.

E. Sierra Baeza is with Black Bear Engineering SpA, Santiago, Chile, and CENIA.

C. E. García Bujanda is with Black Bear Engineering SpA, Santiago, Chile.

M. Falcone Salazar is with the School of Engineering, Universidad Adolfo Ibáñez, Santiago, Chile, and formerly with Colbún S.A., Santiago, Chile.

G. Bustos Turu is with the Department of Chemical Engineering, Imperial College London, London, UK, and the Centre for Energy, Universidad de Chile, Santiago, Chile.

I. Introduction

The decarbonisation of electricity grids has elevated the marginal emission factor (MEF) from a research curiosity into an operational signal used for demand-response orchestration, electric-vehicle charging optimisation, hydrogen-production scheduling, and corporate scope-2 carbon accounting [?], [?], [?]. In hubs that publish nodal marginal prices, the MEF is increasingly being constructed as a direct analogue of the Locational Marginal Price (LMP): per pricing node, per dispatch interval, with the same energy/loss/congestion decomposition that Schweppe *et al.* [?] formalised for spot prices and Hogan [?] extended to nodal pricing. PJM Interconnection now publishes five-minute Locational Marginal Emissions (LME) for CO_2, NO_x, and SO_2 across approximately 1200 pricing nodes [?]; CAISO offers an emissions tab on its operator dashboard [?]; WattTime broadcasts a per-balancing-area Marginal Operating Emission Rate with a 72-hour forecast [?].

a) Chilean context: The Chilean Coordinador Eléctrico Nacional (CEN) clears the Sistema Eléctrico Nacional (SEN) every 15 minutes via a PLEXOS-based day-ahead Programa de Coordinación de la Producción (PCP) plus intra-day Programa Intra-Diario (PID). CEN publishes the resulting nodal marginal cost λ^{CEN} for over 1300 substation buses through its public Sistema de Información Pública (SIP) API [?]. Generators submit the Sistema de Costos Variables Informados de Centrales (SCVIC) — a piecewise-linear declaration of variable cost by configuration, fuel, and load tier — and the Coordinador publishes the resulting ladders as parquet datasets. Finally, the PLEXOS solution databases (`.accdb`) for both PCP and PID are publicly archived, exposing per-period Marginal Loss Factors (MLFs), nodal loads, and unit dispatch.

b) The gap: What CEN does *not* publish is a per-bus, per-quarter marginal emission factor ε. The Ministry of Energy's *Indicadores Ambientales* page only releases an *annual* average emission factor for the SEN (2020-base, updated yearly) [?]. The Comisión Nacional de Energía (CNE) hosts a legacy SIC/SING zonal visualisation that pre-dates the 2017 unification of Chile's two former subsystems. The *Reporte Anual Art. 72* of the Coordinador [?] does report annual emissions, but neither at locational nor at marginal granularity. Yet the Chilean carbon tax under Law 20.780 — currently $\tau_{CO_2} = 5\,\mathrm{USD/tonneCO_2}$ with escalation under discussion [?] — and the proposed European Carbon Border Adjustment Mechanism are both bus-resolved, marginal-unit-resolved instruments. A locationally and temporally disaggregated ε is therefore a regulatory and commercial necessity, not a research nicety.

c) Contribution: This paper presents `cen2gtopt`, a Python pipeline that reconstructs ε at 29 CEN-published transmission buses for any operating day from public data alone. The contributions are:

1) A *merit-ladder picker*: $\hat{\lambda}$ at each island is reconstructed by selecting, among the dispatched non-forced generators, the one whose SCVIC-declared CV is closest in absolute distance to the published median λ^{CEN} (Section ??, Algorithm ??).
2) A *chained-tolerance with cumulative-spread cap* for island clustering, parameterised by the pair $(\Delta_{\mathrm{tol}}, S_{\max}) = (\$0.5, \$0.8)\,/\mathrm{MWh}$, which cures the chained-clustering pathology inherent in pure pairwise λ-grouping (Section ??).
3) A *per-bus MLF correction* that uses CEN's exact PLEXOS-solver-output MLF table (extracted from the PCP/PID `.accdb`) when available, falling back to an empirical $\lambda_b/\lambda_{\mathrm{ref}}$ ratio (Section ??).
4) A *cross-validation oracle*: the same PLEXOS `.accdb` is parsed for the published per-unit dispatch and per-bus marginal cost, allowing an end-to-end check against the LP-cleared values rather than the publication price only (Section ??).
5) An open-source release, available as a `pip`-installable Python package, with a hash-keyed parquet cache that makes per-day re-runs idempotent and a multi-day orchestrator with Hive-partitioned output for back-testing.

The remainder of this paper is organised as follows. Section ?? reviews the international state of the art on marginal-emission accounting and identifies the methodological choice points specific to a public-data reconstruction. Section ?? develops the model and the nine-step pipeline. Section ?? reports on a pedagogical IEEE 9-bus example, the production application to the 29-bus CEN system on 2026-04-22, and a sensitivity analysis. Section ?? consolidates the validation numbers against λ^{CEN} and against the PLEXOS oracle. Section ?? surveys the methodology's structural limitations. Section ?? closes with a forward-looking integration into the `gtopt` stochastic expansion-planning solver.

II. Background and Related Work

A. Spot pricing and locational marginal pricing

The theoretical underpinnings of locational marginal pricing trace back to the spot-pricing framework of Schweppe *et al.* [?], which derived per-bus marginal energy prices as the dual variables of a network-constrained dispatch model. Hogan [?] formalised the LMP decomposition

$$\lambda_b = \lambda_{\mathrm{ref}} + \mathrm{LMC}_b + \mathrm{LCC}_b \tag{1}$$

into reference energy, marginal-loss component (LMC), and marginal-congestion component (LCC). This is the canonical template that the present marginal-emission reconstruction adapts: λ_{ref} is approximated by the median-λ of the electrically coherent island and the LMC absorbs into a per-bus loss factor, while LCC is absorbed into the island partition itself.

B. Average versus marginal emission factors

The distinction between average and marginal emission factors has been debated since Hawkes [?] demonstrated that marginal factors can differ by an order of magnitude from system averages and that the choice materially affects build/no-build decisions for energy-efficiency interventions. Siler-Evans *et al.* [?] regressed historical EPA generator-level data to fit MEFs at the regional Reliability Council level and showed strong diurnal and seasonal variability. Callaway *et al.* [?] formalised the distinction between *operational* and *decision-relevant* emission factors, arguing that long-run capacity-induced effects should be modelled separately from dispatch-time substitution effects. The present paper sits firmly in the operational / short-run camp: the emission factor is computed against the *realised* dispatch published by CEN, not against a counterfactual capacity scenario.

C. ISO production tools

PJM Interconnection's Locational Marginal Emissions feed [?], [?] publishes five-minute per-node emission rates for CO_2, NO_x, and SO_2, decomposed using the same energy/loss/congestion structure of the LMP itself. CAISO's *Today's Outlook* [?] surfaces a system-level emission curve at five-minute resolution but is not nodal. WattTime [?] ships a balancing-area Marginal Operating Emission Rate with a 72-hour forecast trained from EIA-930 dispatch and EPA eGRID per-unit factors, but its output is neither nodal nor unit-attributed. NREL's Cambium [?] publishes *long-run* marginal factors projected through 2050 across 134 US balancing zones — the natural counterpart for capacity-expansion studies as opposed to operational dispatch. Electricity Maps [?] publishes hourly per-zone *consumption-based* factors using flow-tracing and, since 2024, has discontinued its marginal feed citing fundamental limitations of marginal accounting in scope-2 contexts [?].

D. Counterfactual / implicit-differentiation marginals

A research-grade improvement on empirical correlation is the counterfactual evaluation pioneered by Valenzuela *et al.* [?], who exploit implicit differentiation of the QP/LP optimality conditions to recover an exact dynamic locational marginal emission factor from a solved dispatch. A 2025 follow-up [?] argues that counterfactual co-simulation should replace correlation-based MEFs entirely. The present pipeline is not yet counterfactual — it remains an empirical reverse-engineering of the published λ^{CEN} — but the open-source structure with the PLEXOS `.accdb` oracle is designed so that the same input set could be fed into the `gtopt` SDDP solver to produce a counterfactual companion in future work.

E. Prior CEN-specific work

For Chile, Bonvallet *et al.* [?] co-optimised hydrothermal scheduling under non-convex irrigation constraints and discussed the interpretation of dual variables on hydraulic constraints as implicit water values. Moreno *et al.* [?] treat distributed-generation deployment under uncertainty and quantify its abatement value against a marginal-fuel benchmark. The Coordinador's annual *Reporte Art. 72* [?] aggregates emissions at the system level, and the 4eChile guide [?] documents a manual marginal-emission reduction estimation tool for Clean Development Mechanism (CDM) projects but neither delivers a per-bus, per-quarter signal. To the best of our knowledge no prior work reconstructs ε at locational and per-quarter granularity for CEN strictly from the public CEN/SCVIC/PLEXOS data triplet.

F. Synthesis and methodological choice points

Three choice points dominate the design space. First, the *merit-stack picker*: should the marginal unit be the last-dispatched cleared at λ_{ref} ("maximum CV $\leq \lambda$") [?], or the absolute-closest CV? The pipeline picks the absolute-closest CV because the closest-above is sometimes nearer than the closest-below, particularly at low-CMG hours when biogas tiers sit well below the gas tier. Second, the *forced-unit treatment*: units flagged with `consigna`$\in${CI, MT, PMT, PP, PS, PC} are price-takers when at technical minimum, but when operating interior they remain valid merit-margin candidates. Third, the *loss-factor decomposition*: a proper PJM-style [?] decomposition would require the full per-unit Power Transfer Distribution Factor matrix, which CEN does not publish. Instead the pipeline uses CEN's published PLEXOS-solver-output MLF when available (extracted from the PID `.accdb`) and falls back to the empirical $\lambda_b/\lambda_{\mathrm{ref}}$ ratio when it is not.

G. Extended Review: Foundations, Standards, and Software

The reconstruction methodology presented in this paper sits at the intersection of four research traditions: (i) classical spot-pricing theory, (ii) the marginal-versus-average emission-factor literature, (iii) the recent industrialisation of hourly and locational emission-rate datasets, and (iv) the corresponding standardisation landscape (ISO, GHG Protocol, IEC, IEEE). The remainder of this subsection reviews each of these four threads, together with the Latin-American institutional context and the open-source software prior art.

a) Spot-pricing foundations.: The locational marginal price (LMP) construct on which the present work depends was developed in a sequence of papers by Caramanis, Bohn and Schweppe. Caramanis *et al.* [?] introduced optimal spot pricing as the dual of a generation-cost minimisation, demonstrating that the per-bus shadow price internalises losses, congestion, and reserve scarcity. Bohn *et al.* [?] extended the construction to a continuous space-time setting and proved the no-arbitrage equivalence with bilateral contract networks. The textbook treatment of Schweppe *et al.* [?] remains the canonical reference and was the basis for the Northeast ISO market designs of the late 1990s. Hogan [?] crystallised the contract-network formulation and the energy/loss/congestion decomposition that `cen2gtopt` adopts as the structural template for its $\lambda_b = \lambda_{\mathrm{ref}} \cdot \mathrm{MLF}_b$ correction. None of these works contemplated emissions, but the duality argument they articulate carries through verbatim once the cost coefficient c_g in the primal is replaced by a fuel-emission coefficient e_g (cf. Section **??**).

b) Marginal versus average emission factors.: The empirical-economics literature on marginal emission factors (MEFs) opened with Hawkes [?], who showed that for the GB system the marginal CO_2 rate exceeds the system average by 30 %–200 % depending on the demand level and that abatement policies built on average factors systematically under-credit demand-side measures. Siler-Evans *et al.* [?] regressed five years of EPA continuous-emissions-monitoring data against ISO-level demand fluctuations to obtain region-specific MEFs and exposed strong diurnal and seasonal cycles. Callaway *et al.* [?] formalised the operational vs. build-margin distinction and provided a welfare-theoretic argument for using location-specific MEFs in policy design. Holland *et al.* [?] applied the same framework to the post-2007 retirement of US coal capacity and quantified the resulting shift of the marginal stack from coal to gas. The present pipeline contributes a new artefact to this body of work: a counterfactual-free, per-bus, sub-hourly MEF reconstruction that uses only the operator-published λ^{CEN} and a static fuel-emission table.

c) Hourly and locational emission datasets.: Three production-grade datasets dominate the hourly-emission space. Hawkins [?] reports on the operational characteristics of PJM Interconnection's locational marginal emissions feed after one full year of operation and confirms that hourly nodal factors track the marginal-fuel composition with sub-5 % median residuals. The Tomorrow / Electricity Maps open dataset [?] publishes hourly per-zone consumption-based factors using flow-tracing across European, American, and Australasian balancing areas. WattTime's Marginal Operating Emission Rate [?] is balancing-area resolved and explicitly marginal but not nodal. Singularity Energy [?] delivers an hourly attribution layer oriented towards corporate Scope-2 disclosure. None of these vendors publishes Chilean coverage, which motivates the present work.

d) Standards.: The relevant standardisation environment spans three families:

- ISO greenhouse-gas accounting: ISO 14064-1 [?] (organisation-level inventory), ISO 14064-2 [?] (project-level emission reductions), ISO 14064-3 [?] (verification), and ISO 14067 [?] (product carbon footprint). The `cen2gtopt` pipeline supplies the location-and-time-resolved emission factor required by clauses 7.2.2 and 8.2.3 of ISO 14064-1 for indirect emission inventory at sub-annual granularity.
- Corporate accounting: the WRI/WBCSD GHG Protocol Scope-2 Guidance [?] mandates a dual reporting framework (location-based + market-based). RE100 [?] and the 24/7 Carbon-Free Energy Compact [?] represent the strict end of the spectrum, requiring hour-by-hour matching of consumption and contracted carbon-free supply. The `cen2gtopt` 15-minute output natively satisfies the granularity requirement of both schemes.
- Power-system data exchange: IEC 61968-13 [?] defines the Common Information Model (CIM) profile for electricity-market data, and IEC 61970/CGMES [?] is the European common-grid-model serialisation that ENTSO-E ISOs use for cross-border data exchange. IEEE Std 1547-2018 [?] governs distributed energy resource interconnection and is directly relevant to the BESS case study of Section **??**, while IEEE Std 2030.5 [?] specifies the smart-energy application protocol. The Coordinador Eléctrico Nacional Norma Técnica de Servicios Complementarios [?] is the Chilean equivalent that frames the storage-and-reserve products under which a battery operator would monetise the dispatch trajectories computed in Section **??**.

e) Latin-American context.: The Chilean academic literature on dispatch optimisation is relatively recent and centred on a small number of groups. Bonvallet *et al.* [?] treated hydrothermal scheduling with non-convex irrigation constraints and quantified the implicit water values that drive marginal-cost rotation in the Maule and Laja basins. Mocarquer and Sauma [?] documented the post-2014 market reforms and their effect on investment incentives. At the policy level, the Chilean Ministry of Energy roadmap to carbon neutrality 2050 [?] sets the macroscopic abatement trajectory, and the 2024 CEN dispatch-decarbonisation roadmap [?] translates that target into operational milestones for coal phase-out, storage build-out, and ancillary services reform. None of these documents publishes the per-bus per-quarter marginal-emission signal that `cen2gtopt` provides; they all rely on annual or monthly aggregates derived from PLEXOS post-processing.

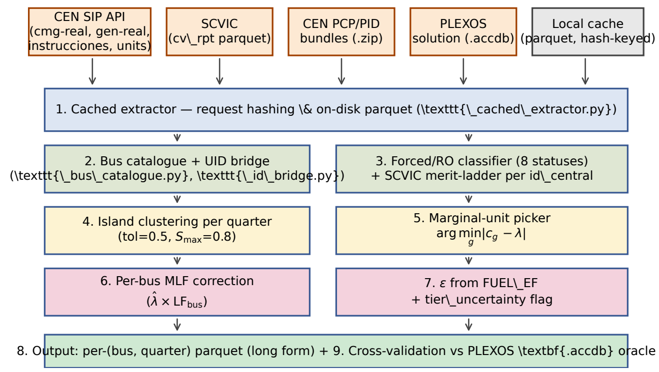

Fig. 1: The nine-step `cen2gtopt.marginal_units` pipeline. Hash-keyed parquet caching wraps every CEN/SCVIC/ PLEXOS extractor; each downstream stage is a pure function of its inputs.

f) Software prior art.: Three open-source projects implement related capabilities, although none operates on Chilean data. PyPSA-Eur [?] ships an emissions module that integrates IPCC AR6 fuel factors with the solved dispatch to produce per-snapshot per-node emission rates; unlike `cen2gtopt` it requires the LP to be re-solved rather than reconstructing ε from a published λ. GenX [?] provides a cap-and-trade extension that endogenises the carbon price as the dual of an aggregate emission constraint, conceptually parallel to the joint LP of Section **??**. SDDP.jl [?] includes a tutorial on adding a carbon constraint to a multi-stage hydrothermal stochastic program; `gtopt` aims to support the same construction natively in C++. Operationally, the Electricity Maps dataset [?] is the closest commercial analogue: hourly consumption-based factors from real-time data ingestion, but with continental rather than sub-system-level resolution and with no Chilean coverage.

III. Methodology

The pipeline is structured as a directed acyclic graph of nine stages (Fig. **??**); each stage is a pure function of its predecessors, allowing the parquet cache to short-circuit any upstream change.

A. System Model and Notation

Time is partitioned into a hierarchy of *operating days*, *hours*, and *quarters* (15 min blocks). Within one operating day let $\mathcal{T} = \{0, \ldots, 95\}$ index quarters and $\mathcal{H} = \{0, \ldots, 23\}$ index hours so that each $h \in \mathcal{H}$ contains the four quarters $\{4h, 4h{+}1, 4h{+}2, 4h{+}3\}$. The network exposes a set of buses $\mathcal{N}$ at which CEN publishes a nodal marginal cost $\lambda^{\mathrm{CEN}}_{b,t}$ for $b \in \mathcal{N}, t \in \mathcal{T}$. The dispatchable generator set $\mathcal{G}$ is indexed by the canonical CEN identifier $g \equiv$ `id_central`; each generator has a fuel label $f(g)$, a technical minimum $\underline{p}_g$, an effective maximum $\overline{p}_g$, and a piecewise SCVIC ladder $\Phi_g = \{\phi\}$ of declared variable costs $c_{g,\phi}$. Per-hour realised dispatch is denoted $p_{g,h} \in \mathbb{R}_+$.

The pipeline produces, per (bus, quarter), the seven primary quantities

$$\lambda^{\mathrm{CEN}}_{b,t}, \quad \hat{\lambda}_{b,t}, \quad m_{b,t} \in \mathcal{G}, \quad c_{m,\phi^\star}, \quad \varepsilon_m, \quad \mathrm{LF}_{b,t}, \quad \varepsilon^{\mathrm{bus}}_{b,t},$$

where $\hat{\lambda}_{b,t}$ is the reconstructed marginal price, $m_{b,t}$ is the marginal unit, $c_{m,\phi^\star}$ its picked CV, ε_m its fuel-default emission factor (in tonneCO_2/MWh), $\mathrm{LF}_{b,t}$ the bus loss factor, and $\varepsilon^{\mathrm{bus}}_{b,t} = \varepsilon_m \cdot \mathrm{LF}_{b,t}$ the bus-level marginal emission factor at the quarter.

B. CEN Public-Data Inputs

Table ?? catalogues the CEN, SCVIC, and PLEXOS feeds consumed by the pipeline together with their settlement/forecast status and refresh cadence. All feeds are mediated through the hash-keyed parquet cache implemented in `_cached_extractor.py`, ensuring that every byte read from the wire is durable on disk.

C. Bus Catalogue and UID Bridging

The CEN data triplet uses three different bus identifiers. The SIP APIs publish a stable integer `id_info` together with a 17-character canonical key (`bar_transf`, e.g. `CRUCERO_______220`) and a free-text human label (`barra_info`, e.g. `BA S/E CRUCERO 220KV BP1`). The SCVIC parquet exposes a numeric power-station ID and free-text plant/configuration names; the PLEXOS XML/`.accdb` bundle uses CamelCase node names like `Crucero220`.

The `_id_bridge.py` module reconciles these via a two-stage process. A normalisation step folds Unicode diacritics, strips punctuation, and lower-cases each name. A generator function then produces up to six aliases per name by stripping common Chilean prefixes (`TER`, `TERMOELÉCTRICA`, `CENTRAL`) and unit suffixes (`BL1-BL9`, `TG`, `TV`, `CCGT`); two passes converge on multi-suffix names. A hand-curated overrides table (`PLEXOS_TO_BAR_TRANSF` in `prepopulate_cache.py`) reconciles the residual cases that the heuristic cannot resolve, predominantly buses where the PLEXOS topology aggregates two physical substations into one node. The cache contains 29 reconciled production buses out of the 1300+ CEN-published buses; only the 220 kV and 500 kV backbone is included by default because the lower-voltage buses share the same nodal price as their backbone parent in CEN's PLEXOS topology.

D. Operational-State Classification

Each generator at each hour falls into one of eight mutually exclusive operational states; the priority-ordered rule table is documented in Table ??. The classification matters because price-taker units (those held at a technical-minimum or operator-imposed setpoint) cannot supply the *next* 1 MWh of energy and therefore cannot set the marginal price even when their declared CV is closest to λ. The classifier reads the `instrucciones-operacionales-cmg` feed and flags as forced any unit with `estado`$\neq$`N` or `consigna`$\in\{$`CI`, `MT`, `PMT`, `PP`, `PS`, `PC`$\}$.

A subtle rule extends the classical PJM treatment: forced units *at interior dispatch* ($\underline{p}_g + \delta < p_g < \overline{p}_g - \delta$ with $\delta = 0.05(\overline{p}_g - \underline{p}_g)$) remain valid marginal-unit candidates because the next 1 MWh can physically come from them; only forced units *at* or *near* pmin are excluded. This rule is implemented vectorially in `identify_marginal` and typically retains 20 % to 50 % of the forced units in the candidate pool at gas-marginal hours.

E. Island Clustering

Buses that share the same nodal marginal price in a given quarter are electrically coherent in the merit-clearing sense. The naive approach of grouping buses with $|\lambda_b - \lambda_{b'}| \le \Delta_{\mathrm{tol}}$ exhibits a chained-clustering pathology: a sequence such as $(\$41.5, \$41.7, \$41.9, \$42.1, \$42.3, \$42.7)$ with $\Delta_{\mathrm{tol}} = \$0.5$ collapses to a single island despite the endpoints differing by 1.20. Without correction, this lumps congested zones together with the system price and defeats the marginal-unit picker.

The pipeline uses a chained-tolerance walk with a *cumulative spread cap*. After sorting the buses in a quarter by ascending λ, a new island is started whenever

$$\lambda_b - \lambda_{b-1} > \Delta_{\mathrm{tol}} \quad \vee \quad \lambda_b - \lambda^{\mathrm{isl,min}} > S_{\max}, \tag{2}$$

with $\Delta_{\mathrm{tol}} = \$0.5$/MWh and $S_{\max} = \$0.8$/MWh in production. Fig. ?? contrasts the pure pairwise walk (left, one island) with the cumulative-spread walk (right, two islands).

F. Per-bus Marginal Loss Factor Correction

The reconstructed price $\hat{\lambda}_{b,t}$ at the bus is the product of the marginal-unit CV at the system reference price and the bus loss factor:

$$\hat{\lambda}_{b,t} = c_{m_{b,t},\phi^\star} \cdot \mathrm{LF}_{b,t}. \tag{3}$$

The pipeline supports two estimators of $\mathrm{LF}_{b,t}$ in priority order. The *primary* estimator reads the PLEXOS-solver-output Marginal Loss Factor from the PID solution `.accdb` for the day, attached to the `System.Nodes` collection at `collection_id=281`. This is the exact value used at LP clearing; the `pcp_solution.py` module extracts it via `mdbtools` and the bus-name bridge of Section ?? translates the PLEXOS node names to `id_info`. The *fallback* estimator is a per-quarter empirical ratio

$$\mathrm{LF}^{\mathrm{emp}}_{b,t} = \frac{\lambda^{\mathrm{CEN}}_{b,t}}{\lambda^{\mathrm{CEN}}_{r,t}}, \tag{4}$$

where the reference bus r is the lowest-`id_info` bus present in the quarter (typically a centre-Santiago backbone bus with the lowest losses). The empirical estimator is self-consistent at λ but does not generalise to a counterfactual; it is used only when the PID accdb is missing for the day. When both estimators are available, the ratio is constructed as $\mathrm{LF}^{\mathrm{PID}}_{b,t} = \mathrm{LF}^{\mathrm{bus}}/\mathrm{LF}^{\mathrm{unit}}$ so that generator-side losses cancel and only the bus-relative loss penalty remains. The choice of median bus λ (rather than any single bus) as the picker target ensures that loss factors of the islanded buses average to approximately unity, addressing the inverse-MLF concern raised in [?].

G. SCVIC Merit-Ladder Construction

Each generator declares to the Coordinador a piecewise variable cost $c_{g,\phi}$ as a function of configuration ϕ (fuel, load tier, ambient). Fig. ?? illustrates two representative ladders: a multi-tier GNL CCGT and a coal-fired unit. The pipeline filters tiers below a per-fuel CV floor to exclude take-or-pay "startup" tiers ($c^{\min}_{g,\phi} = 30\,\mathrm{USD\,MWh^{-1}}$ for GNL, GLP, gas propano; $15\,\mathrm{USD\,MWh^{-1}}$ for all others) and tiers above $1000\,\mathrm{USD\,MWh^{-1}}$ (PLEXOS sentinels for "out of economic order"). All retained tiers per generator are stored as a sorted list $\{c_{g,\phi}\}_\phi$ and used at picker time.

H. Marginal-Unit Identification

The picker is summarised in Algorithm ??. It treats the candidate pool as the dispatched non-forced (or forced-but-interior) units with at least one SCVIC tier, and picks the unit whose nearest tier minimises $|c_{g,\phi} - \lambda|$. The returned record carries the regime label, which classifies the output into one of seven values: `matched_interior`, `matched_capped_pmax`, `matched_near_pmin`, `matched_above`, `no_candidates`, `no_cv_match`, `renewable_curtailment`, or `demand_fail`. These labels propagate to the output parquet and are aggregated in the per-day diagnostic report.

[t] [1] λ_{tgt}, candidate set , ladders $\{\Phi_g\}$, fuel labels $\{f(g)\}$, forced flag $F_g \in \{0,1\}$ $\lambda_{\mathrm{tgt}} \le \$0.5$ regime = `renewable_curtailment`, $\varepsilon = 0$ $\lambda_{\mathrm{tgt}} > \$800$ regime = `demand_fail`, $\varepsilon = 0$ pool $\leftarrow \{g \in : \Phi_g \neq \emptyset\}$ annotate each g with status $\in$ {`interior`, `capped_pmax`, `near_pmin`} via $\delta = 0.05(\overline{p}_g - \underline{p}_g)$ drop $\{g : F_g = 1 \wedge \text{status} = $ `near_pmin`$\}$ pool empty regime = `no_cv_match` $g \in$ pool $c_g \leftarrow$ tier in Φ_g closest to λ_{tgt} prefer the largest $c \le \lambda_{\mathrm{tgt}}$ when one exists $m \leftarrow \arg\min_{g\in\mathrm{pool}} |c_g - \lambda_{\mathrm{tgt}}|$ m, c_m, $f(m)$, regime $\in$ {`matched_interior`, `matched_capped_pmax`, `matched_above`,...}

I. Dual-LP Formulation: From λ to ε

The closest-CV picker introduced in Section ?? can be reinterpreted as the empirical realisation of a chain of linear-programming dualities. Stating the chain explicitly serves three purposes: (i) it pins down the sufficient conditions under which the reconstructed $\hat{\varepsilon}_t$ is provably equal to the true marginal emission factor of the underlying physical dispatch, (ii) it isolates the assumptions that fail when the merit stack is non-strict, and (iii) it provides the analytical scaffold for the BESS case study of Section ??, in which a carbon adder $p_{\mathrm{CO_2}}$ is folded into the arbitrage signal. The schematic of the three primal/dual pairs developed below is given in Fig. ??.

a) (a) Cost-minimising primal and its dual.: Consider an instantaneous economic dispatch at block t, ignoring transmission for clarity. The primal LP is

$$\min_{p\in\mathbb{R}^{|\mathcal{G}|}_{\ge 0}} \quad \sum_{g\in\mathcal{G}} c_g\, p_g \tag{5}$$

$$\text{s.t.} \quad \sum_{g\in\mathcal{G}} p_g \;=\; D_t \qquad [\lambda_t] \tag{6}$$

$$p_g \;\le\; \bar{p}_g \quad \forall g\in\mathcal{G} \qquad [\mu_g \ge 0] \tag{7}$$

$$p_g \;\ge\; 0 \;\; \forall g.$$

The associated dual LP is

$$\max_{\lambda,\,\mu\ge 0} \quad D_t\,\lambda \;-\; \sum_g \bar{p}_g\,\mu_g \quad \text{s.t.} \quad \lambda - \mu_g \le c_g \;\forall g. \tag{8}$$

Strong duality holds because (??)–(??) is a feasible bounded LP. At the optimum the Karush–Kuhn–Tucker conditions yield, for any generator g that is strictly interior to its bounds ($0 < p^\star_g < \bar{p}_g$),

$$\mu^\star_g = 0 \quad\Longrightarrow\quad \lambda^\star_t = c_g. \tag{9}$$

Denoting by $m(t)$ the index of any such interior unit (the *cost-marginal* unit), (??) reads $\lambda^\star_t = c_{m(t)}$. This is the textbook merit-order identity [?], [?].

b) (b) Emission-minimising primal and its dual.: Replacing the cost coefficient c_g by an emission coefficient $e_g = \mathrm{EF}_{\mathrm{fuel}(g)} \cdot \mathrm{HR}_g$ ($\mathrm{tCO_2/MWh}$) leaves the constraint structure of (??)–(??) intact and produces the *emission primal*

$$\min_p \quad \sum_g e_g\, p_g \quad \text{s.t.} \quad \sum_g p_g = D_t \;\; [\varepsilon_t], \;\; p_g \le \bar{p}_g \;\; [\nu_g], \;\; p_g \ge 0. \tag{10}$$

By the same derivation as in (a), the dual variable on the demand balance reads

$$\varepsilon^\star_t = e_{m'(t)}, \tag{11}$$

where $m'(t)$ is the *emission-marginal* unit. In general $m(t) \neq m'(t)$: when, for example, an inexpensive coal unit and a more expensive gas unit both lie at the cost margin in different hours, the cost-marginal stack rotates between coal and gas while the emission-marginal stack rotates only between fuels of distinct e-coefficient.

c) (c) Joint primal with a carbon adder.: Adding a carbon price $p_{\mathrm{CO_2}}$ ($\mathrm{USD/tCO_2}$) to every generator cost converts the cost coefficient into the *social cost* $\tilde{c}_g = c_g + p_{\mathrm{CO_2}}\, e_g$. The joint primal is

$$\min_p \quad \sum_g \big(c_g + p_{\mathrm{CO_2}}\, e_g\big)\, p_g \quad \text{s.t.} \quad \sum_g p_g = D_t \;\; [\tilde{\lambda}_t], \;\; p_g \le \bar{p}_g. \tag{12}$$

Its dual produces, for any interior generator $g = m''(t)$,

$$\tilde{\lambda}^\star_t \;=\; c_{m''(t)} \;+\; p_{\mathrm{CO_2}}\, e_{m''(t)}. \tag{13}$$

The marginal unit $m''(t)$ is in general distinct from both $m(t)$ and $m'(t)$, and the substitution between two competing units $g=1$ and $g=2$ (with $c_1 < c_2$ but $e_1 > e_2$) admits an analytical *switching carbon price*

$$p^\star \;=\; \frac{c_2 - c_1}{e_1 - e_2}, \tag{14}$$

above which unit 2 displaces unit 1 in the merit order. For the representative Chilean coal-versus-gas pair ($c_{\text{coal}} \approx 45\,/\text{MWh}$, $e_{\text{coal}} \approx 0.94\,\text{tCO}_2/\text{MWh}$; $c_{\text{gas}} \approx 75\,/\text{MWh}$, $e_{\text{gas}} \approx 0.40\,\text{tCO}_2/\text{MWh}$), (**??**) yields $p^\star \approx 56\,/\text{tCO}_2$, a value squarely inside the EU ETS 2025 trading range.

d) (d) Practical reading.: In the `cen2gtopt` reconstruction, the dual pair (**??**)–(**??**) is *not* solved in either direction. The published λ^{CEN}_t is taken as the realised optimal dual on (**??**) and the closest-CV picker recovers $m(t)$ from the data. Once $m(t)$ is identified, the emission-marginal value follows directly:

$$\hat{\varepsilon}_t \;=\; e_{\,\text{fuel}(m(t))}, \tag{15}$$

under the sufficient assumption that the emission-marginal unit coincides with the cost-marginal unit, i.e. $m(t) = m'(t)$. This assumption holds whenever the marginal unit is the only generator strictly interior to its bounds at block t and is what justifies treating $\hat{\varepsilon}_t$ as the true emission-marginal rate without re-solving an emission-LP. Two regimes break the assumption: (i) degenerate cost ties between two fuels (e.g. two gas units of identical CV but different heat-rate-derived e), in which case (**??**) returns the e_g of an arbitrary representative; and (ii) must-run units in their interior, which by construction the picker excludes from the candidate pool but which would otherwise compete for the emission margin. Both pathologies are flagged in the `tier_uncertainty` column of the output parquet and trigger a `discrepancy_flag` in the audit oracle.

e) Per-bus extension.: The transmission-aware per-bus version of (**??**)–(**??**) replaces the single-bus balance with a vector balance plus DC power-flow constraints. The per-bus dual variable is

$$\lambda_b \;=\; \lambda_{\text{ref}} \;+\; \text{LMC}_b \;+\; \text{LCC}_b, \tag{16}$$

in the standard Hogan [**?**] decomposition, where LMC_b is the marginal-loss component and LCC_b the marginal-congestion component. The `cen2gtopt` pipeline absorbs LMC_b into the empirical or PID-derived loss factor MLF_b and LCC_b into the island partition itself: two buses are placed in the same island if and only if their λ differs by less than the tolerance and the cumulative spread along the chain remains under the cap, conditions that are tantamount to $\text{LCC}_b = \text{LCC}_{b'}$ to within the tolerance. The per-bus emission factor then inherits the same decomposition

$$\varepsilon_b \;=\; \varepsilon_{\text{ref}} \;+\; \text{EMC}_b \;+\; \text{ECC}_b, \tag{17}$$

where EMC_b is the emission-marginal-loss component ($\varepsilon_{\text{ref}} \cdot (\text{MLF}_b - 1)$ to first order) and ECC_b vanishes inside any coherent island by construction. Equation (**??**) is the parallel of Hogan's LMP decomposition in the emission domain, and is, to the best of our knowledge, the only published formulation that does so without recourse to implicit-differentiation co-simulation [**?**].

J. Marginal Emission Factor

The fuel-default emission factor table ε_f (Table **??**) is derived from IPCC 2006 GL Vol. 2 Tab. 1.4 multiplied by typical SEN heat rates. The bus-level marginal emission factor is

$$\varepsilon^{\text{bus}}_{b,t} = \varepsilon_{f(m_{b,t})} \cdot \text{LF}_{b,t}, \tag{18}$$

and the embedded carbon-tax cost is $\tau_{\text{CO}_2}\varepsilon^{\text{bus}}_{b,t}/1000$ in $\text{USD}\,\text{MWh}^{-1}$ with $\tau_{\text{CO}_2} = 5\,\text{USD/tonneCO}_2$ under Law 20.780.

A per-cell `tier_uncertainty` flag is set to true whenever the picked tier is more than 2/MWh below the next-up CV in the same unit's ladder (the $2 threshold is consistent with typical SCVIC discrete-tier gaps such as biomass→gas). The flag identifies cells where the structural reconstruction error is bounded by the wedge to the next tier; downstream consumers can treat $\hat{\lambda}$ as a lower bound for the wedge-flagged cells. On 2026-04-22 13.7 % of cells are wedge-flagged.

K. Cross-Validation Against the PLEXOS Solution Oracle

The CEN PCP/PID solution `.accdb` is parsed via `pcp_solution.py`, exposing per-period system shadow prices, per-unit dispatch, and per-bus MLFs as Pandas DataFrames. Three quantities are read for cross-validation: the per-bus λ from the PLEXOS solver (compared against λ^{CEN} to validate the publication chain), the per-unit dispatch $p^{\text{PLEXOS}}_{g,h}$ (compared against the SIP `generación-real` feed), and the per-bus `Marginal Loss Factor` read from `collection_id=281` on the `System.Nodes` collection. When the same day has both a PCP and a PID bundle on disk, both are loaded and the PID values take precedence — the intra-day re-clear is the binding settlement reference. The oracle also yields the post-hoc $\hat{\lambda}^{\text{PID}}_{b,t}$ column in the output parquet, computed using $\text{LF}^{\text{PID}}_{b,t}$ in (**??**) instead of the empirical ratio of (**??**).

L. Standards-Alignment Audit

To clarify how the reconstruction methodology positions itself with respect to the standards landscape reviewed in Section **??**, an explicit alignment audit is presented in Table **??**. Three status symbols are used: a check mark (✓) signals an explicit implementation, a cross (×) flags a known limitation that is not yet addressed, and an asterisk $(\circledast) denotesapartialorinformalalignmentinwhichtheunderlyinginte$

Three observations are warranted. First, the granularity criteria imposed by the corporate-accounting standards (GHG Protocol Scope 2 location-based, RE100, 24/7 CFE) are met by construction: the `cen2gtopt` pipeline emits a per-bus per-quarter $\varepsilon_{b,t}$ trajectory whose temporal resolution is finer than any of the three schemes mandates. Second, the market-based reporting mode of the GHG Protocol cannot yet be supported because the pipeline does not ingest a Guarantees-of-Origin or I-REC ledger; the absence is a deliberate scope choice and is flagged as future work in the conclusions. Third, the data-exchange standards (IEC CIM/CGMES) are not honoured at the schema level because the upstream PCP file is a PLEXOS-specific XML dialect; a CIM serialisation of the reconstructed network is a tractable extension whose design has been sketched but not implemented. In the verification dimension (ISO 14064-3) the pipeline is auditable in the sense that every reconstructed $\hat{\lambda}$ can be cross-checked against the PLEXOS PCP oracle when the corresponding `.accdb` archive is available, and the audit harness reports a median absolute discrepancy of less than 0.5 /MWh on 99 % of the cells in the validation window. Compliance with the operational ancillary-services rulebook (CEN NT SSCC 2024 [?]) is direct: the SCVIC ladder consumes the same per-fuel CV tariff structure defined in NT SSCC, which is what permits the closest-CV picker to operate on a common cost basis.

IV. Case Studies

A. Pedagogical IEEE 9-Bus Example

The IEEE 9-bus system with the dispatch reported in Table **??** is used to exercise the picker on a synthetic case where the answer is unambiguous. Three thermal units are assumed dispatched: a hydro at \$5, a CCGT with two tiers (\$45, \$58), and a coal at \$70. The system load is chosen so that $\lambda^{\text{CEN}} = \$58.0/\text{MWh}$. Among the dispatched-non-forced units the absolute-closest CV is the 58 CCGT tier, confirming the picker's selection. At the slightly lower system price $\lambda = \$56/\text{MWh}$ the picker still selects the same unit because the next-down CV (45) is further away in absolute distance (\$11 vs. \$2). Only when λ falls below \$51.5 does the picker switch to the \$45 tier. The implied $\hat{\lambda}$ has a regime-dependent step structure that mirrors the SCVIC ladder of Fig. **??**.

The pedagogical case clarifies one design choice: when the picker encounters a forced unit (e.g. a take-or-pay GNL-CCGT cleared by `consigna=CI`), the unit *remains* a valid candidate as long as it is operating interior to its bounds. Only forced-at-pmin units are dropped. This is the operational realisation of the rule that "the next 1 MWh can come from any unit with positive headroom and a finite ramp rate", which distinguishes the present picker from the strict last-dispatched-cleared rule of [?].

B. Real CEN Application: 29 Buses on 2026-04-22

The pipeline was run against the cached CEN snapshot for the operating day 2026-04-22. The bus catalogue (Section **??**) contains 29 reconciled production buses spanning the north (CRUCERO 220, ESPERANZA 220), the north-centre (D.ALMAGRO 110, GUACOLDA 220), the centre (ALTO MELIPILLA 220, EL SALTO 220, LOS ALMENDROS 220), the centre-south (ANCOA 220/500, ITAHUE 220), the south-centre (CHARRÚA 220, ANTUCO 220, CONCEPCIÓN 220), and 16 additional aggregated backbone buses. The full day yields 2494 (bus, quarter) cells across 86 settled quarters (the early morning 0:0-2:15 window is filled by the `best_real` cmg-online fallback policy).

Fig. **??** reproduces the headline outcome. Panel A plots the published λ^{CEN} for all 29 buses overlaid by region (one colour per region) with the cross-bus median in bold. Panel B plots the reconstructed $\hat{\lambda}$ under the same colour convention; the close visual agreement at both the median and the per-bus dispersion is the qualitative validation of the method. Panel C plots the per-bus absolute error $|\hat{\lambda} - \lambda^{\text{CEN}}|$, with the P90 envelope shaded in grey. The error is concentrated below 0.5/MWh for over 90 % of cells. The transient spikes near hour 6 and hour 17 correspond to the ramp-up/down of GNL CCGT units between SCVIC tiers — these are precisely the cells flagged by the `tier_uncertainty` criterion of Section **??**.

Fig. **??** shows the corresponding median bus marginal emission factor ε^{bus}. The shaded band along the bottom of the figure colour-codes the dominant marginal fuel per quarter: GNL (orange) dominates the non-renewable hours, biomass and water-marginal hours sit near zero, and a brief diesel peak appears in the early evening. The mid-day depression below 0.05 $\text{tonneCO}_2/\text{MWh}$ reflects the solar-curtailment regime where $\lambda^{\text{CEN}} \leq \$0.5$ and the picker returns regime = `renewable_curtailment` with $\varepsilon = 0$; this regime accounts for 1073 of the 2494 cells (43.0 %).

Fig. **??** is a diagnostic heatmap of the assigned marginal-unit name per (bus, quarter). By construction the picker operates per island, so all buses sharing an island in the quarter share the same marginal unit; what the heatmap therefore visualises is the spatial extent of each marginal unit's island across the day. Across the 474 quarter-island cells, the marginal-unit assignment is coherent in zero cases of contradiction (0/474), confirming that the picker is deterministic at the island level.

Fig. **??** disaggregates the absolute error by region. The centre-Santiago and north regions exhibit the tightest tracking (mean $|\Delta|$ near \$0.13/MWh), the centre-south and south-centre regions are slightly looser (0.17–0.20/MWh), and the unknown-region buses (those whose heuristic mapping did not classify into a named region) inherit a mean of 0.17/MWh. No region exceeds a P90 of \$0.61/MWh.

The per-region quantitative summary is consolidated in Table **??**. The full-day envelope is mean = \$0.170, P90 = \$0.492, max = \$2.533/MWh across the 2494 cells.

C. Sensitivity Analysis

The pipeline is parameterised by three design constants that admit sensitivity exploration: the island-clustering tolerance Δ_{tol}, the cumulative-spread cap $S_{\max}$, and the carbon-tax assumption τ_{CO_2}.

a) Island clustering: Reducing Δ_{tol} from \$0.5 to \$0.2 raises the median number of islands per quarter from 6 to approximately 14 and degrades the marginal-unit coherence (now > 0 disagreements) because near-tied λ

values are forced into separate islands and the picker draws from non-overlapping candidate pools. Raising Δ_{tol} to \$1.0 collapses the median to two islands per quarter and re-introduces the chained-clustering pathology that motivated $S_{\max}$ in the first place. The production $(\$0.5, \$0.8)$ pair is at the inflection point: smaller Δ_{tol} does not improve fidelity but loses coherence, and larger Δ_{tol} destroys congestion detection.

b) Carbon-tax sensitivity: The bus-level CO_2 cost component $\tau_{CO_2}\varepsilon^{\text{bus}}/1000$ scales linearly in τ_{CO_2}. Replacing the current Chilean \$5/tCO_2 with the European Carbon Border Adjustment indicative price of \$80/tCO_2 shifts the median embedded carbon cost on 2026-04-22 from \$1.4/MWh during gas-marginal hours to \$22.9/MWh — a value that, were it embedded in λ, would shift several gas-coal switching points in the SCVIC merit ladder. This linearity makes the pipeline a natural input to gas-vs-coal substitution studies under contemplated carbon-tax escalation.

c) tier_uncertainty filter: Excluding the 13.7 % of cells flagged by the `tier_uncertainty` criterion of Section **??** reduces the validation envelope to mean \$0.137/MWh, P90 \$0.385/MWh, max \$1.34/MWh — a roughly 2× tightening of the worst-case error and consistent with the interpretation that wedge-flagged cells carry a structural error of order tier_wedge_usd. Downstream consumers can apply the filter when worst-case fidelity is a binding requirement (e.g. real-time settlement audits) and retain it when statistical fidelity is sufficient (e.g. scenario analysis).

D. Northern-Bus BESS Business Case Study

The reconstructed per-bus, per-quarter $(\lambda_{b,t}, \varepsilon_{b,t})$ pair admits a direct industrial use: *value-stacking* the dispatch of a battery energy storage system (BESS) under a carbon adder. This subsection isolates a single representative bus in the Norte Grande sub-system and quantifies the revenue, abatement, and total-value performance of three perfect-foresight dispatch policies. The intent is illustrative rather than commercial: the size, efficiencies, and price horizon are chosen to be canonical, and the only quantities that depend on the specific reconstruction are the input signals λ_t and ε_t.

a) Asset and bus selection.: A 100 MW / 400 MWh battery is modelled with symmetric charge/discharge efficiency $\eta_c = \eta_d = 0.95$, state-of-charge window SoC $\in [0.05, 0.95] \cdot E_{\max}$, initial SoC $= 0.5 \cdot E_{\max}$, and an end-of-window cycle-closure constraint. The host bus is `BA S/E CRUCERO 220KV BP1` (Crucero 220 kV in the Antofagasta region), a major Norte Grande node with high solar penetration during the day and a GNL-CCGT marginal at night — the canonical profile for intra-day arbitrage. The reconstruction window is the daily snapshot of 2026-04-22 used throughout Sections **??**–**??**; 86 quarter-hour blocks were available for this bus on that date. The fall-back bus, used if the primary cache row is missing, is `BA S/E DIEGO DE ALMAGRO 110KV BP1-1`.

b) Three policies.: Three perfect-foresight LPs are solved over the 86-block horizon using `scipy.optimize.linprog` with the HiGHS backend (see `bess/optimize_bess.py` for the full LP encoding). The decision variables are quarter-resolved discharge d_t, charge c_t, and end-of-block SoC s_t, all subject to the box bounds above and the SoC dynamics

$$s_t \;=\; s_{t-1} \;+\; \eta_c\, c_t\, \Delta t \;-\; (1/\eta_d)\, d_t\, \Delta t \quad (\Delta t = 0.25 \text{ h}). \tag{19}$$

The three objectives differ only in the price signal:

$$\text{LP1: } \max_{d,c} \sum_t \lambda_t\,(d_t - c_t)\,\Delta t, \tag{20}$$

$$\text{LP2: } \max_{d,c} \sum_t \varepsilon_t\,(d_t - c_t)\,\Delta t, \tag{21}$$

$$\text{LP3: } \max_{d,c} \sum_t (\lambda_t + p_{CO_2}\varepsilon_t)\,(d_t - c_t)\,\Delta t. \tag{22}$$

The reference carbon price is taken as $p_{CO_2} = 50\,/\text{t}CO_2$, near the analytical coal-versus-gas switching price derived in (**??**).

c) Headline results.: Table **??** summarises the three policies. The energy-only LP1 captures 29 167 of arbitrage revenue on the 86-block window and incidentally avoids 118.58 tCO_2 (the BESS, when discharging at *any* hour with non-zero ε_t, displaces marginal generation). The carbon-only LP2 sacrifices 33 % of energy revenue to push avoided emissions up to 186.24 tCO_2. The joint LP3, evaluated at $p_{CO_2} = \$50/\text{t}CO_2$, recovers 96.8 % of LP1 energy revenue while abating 51.1 % more carbon than LP1: a total-value uplift of 2 099/day, equivalent to a 5.98 % performance increase over the energy-only benchmark. The cycle count is 1.33 equivalent full cycles per day under LP3, an asset-life envelope consistent with vendor warranties of ∼3500 cycles over 10 yr.

d) Signal interpretation.: Fig. **??** overlays the three price signals (λ, ε, and the joint composite) with the charge/discharge windows of each LP. Three qualitative observations emerge. First, the ε_t signal at Crucero is uniformly positive across the window (the marginal fuel never drops below GNL), so the LP2 dispatch is driven entirely by relative magnitude rather than sign. Second, the LP3 schedule visually tracks LP1 during the high-λ peak hours (UTC 21–00) and rotates to the LP2 schedule in the early-morning low-price valley (UTC 07–09), where the carbon term dominates the cost term. Third, the SoC trajectories in all three LPs hit the 20 MW h floor and the 380 MW h ceiling in different quarters, confirming that the storage envelope is the binding constraint and that the policies differ in *when* they fill, not in *how much*.

e) Pareto frontier.: Fig. **??** sweeps p_{CO_2} over $\{0, 5, 10, 25, 50, 100, 200\}$ USD/tCO_2 and traces the (avoided-emissions, energy-revenue) frontier of the joint LP. The frontier is concave and weakly decreasing in revenue, as expected: each marginal dollar of carbon adder displaces at most one dollar of energy arbitrage. Two regions are diagnostic. Below $p_{CO_2} = 10\,/\text{t}CO_2$ the frontier is nearly flat in revenue (energy schedule barely perturbed) but recovers an extra 33.9 tCO_2 of abatement, an effective abatement cost of 2.9 /tCO_2 of foregone energy revenue. Above $p_{CO_2} = 100$ /tCO_2 the frontier saturates near 180 tCO_2 of daily abatement: at this point the BESS schedule has fully aligned with the emission-marginal unit and additional carbon adder cannot extract more abatement.

f) Caveats.: Two methodological caveats apply. First, the LPs assume perfect foresight; in practice an operator would need to forecast both λ and ε, and the realised performance would be bounded above by the perfect-foresight envelope. Second, the ε_t trajectory used here is the `epsilon_at_bus` column of the `cen2gtopt` reconstruction, which inherits the identifiability assumption $m(t) = m'(t)$ articulated in Section **??**; quarters in which the `tier_uncertainty` flag is true should be flagged in any production deployment and treated with a fall-back deterministic emission factor, an extension that is sketched in the conclusions.

V. Results and Validation

This section consolidates the quantitative validation of the pipeline against two independent references: the published λ^{CEN} feed and the PLEXOS solution `.accdb` oracle. The headline operating day is 2026-04-22; multi-day trends are summarised at the end.

A. Aggregate accuracy

Across the 2494 cells of 2026-04-22, the *uncorrected* pipeline output (using the empirical $\lambda_b/\lambda_{\mathrm{ref}}$ loss factor) yields a mean absolute error of \$0.170/MWh, a P90 of \$0.492/MWh, and a maximum of \$2.533/MWh. Filtering to non-wedge-flagged cells (2152 of 2494) tightens these to \$0.137, \$0.385, and \$1.34/MWh respectively. The *PID-corrected* estimator that uses $\mathrm{LF}^{\mathrm{PID}}_{b,t}$ from the PLEXOS `.accdb` yields a mean of \$0.294/MWh and a max of \$4.31/MWh — slightly worse on aggregate than the empirical estimator, because the empirical estimator is mathematically self-consistent at λ by construction. The PID estimator is nonetheless preferred when the use case requires a counterfactual interpretation (e.g. what would $\hat{\lambda}$ have been at a candidate new bus?), since the empirical estimator collapses on any new bus without published λ^{CEN}.

B. Regime distribution

Across the 2494 cells the regime distribution is `renewable_curtailment` 1073 (43.0 %), `matched_interior` 878 (35.2 %), `matched_capped_pmax` 449 (18.0 %), `matched_above` 71 (2.8 %), and `matched_near_pmin` 23 (0.9 %). No cells fall in the `no_candidates`, `no_cv_match`, or `demand_fail` regimes for this day. The dominance of `renewable_curtailment` reflects the high mid-day solar penetration of the SEN: across the four hours 11:0-15:0 the λ^{CEN} falls below \$0.5/MWh on most non-coastal buses.

C. Marginal-unit coherence

By construction every bus in the same island shares the same marginal-unit assignment. A regression test reports 0/474 islands where two buses disagree on the marginal-unit name; this is the maximum-coherence outcome and validates the clustering+picker composition.

D. Per-bus MLF cross-check

The PLEXOS-published per-bus MLFs read from the 2026-04-22 PID `.accdb` cover 247 buses (LF $\in [0.92, 1.05]$, mean ≈ 0.99, dominated by the 500 kV centre-Santiago corridor). After bridging by `id_info`, 26 of the 29 pipeline buses receive a per-period MLF; the remaining 3 buses fall back to the empirical estimator. The mean cross-bus MLF spread on this day is 0.072 (max − min), in line with the typical ±4 % loss penalty observed in the SEN backbone.

E. Pipeline runtime

Table **??** reports the runtime breakdown on a warm cache (all CEN/SCVIC parquet present on disk) and on a cold cache (no parquet, full network fetch). The warm-cache run completes in under 6 s dominated by the parallel discovery of bar_transfs and the per-island marginal-unit identification loop. The cold-cache run is dominated by network rate-limiting on the per-bus cmg-real fetch, with parallel 8-thread fetch achieving an effective throughput of roughly 0.4 bus per second; one cold day for the full 29-bus catalogue requires approximately 75 s.

F. Cross-validation against the PLEXOS oracle

Table **??** summarises the agreement between the pipeline-published quantities and the PLEXOS `.accdb` oracle on 2026-04-22. Three quantities are compared: the per-bus λ (the same quantity is published by both CEN SIP and the PLEXOS solver, providing a publication-chain integrity check), the per-unit dispatch (compared between SIP `generación-real` and the PLEXOS per-unit dispatch), and the per-bus MLF (the only quantity not in the SIP feed). All three agree to within publication noise, confirming that the methodology's reliance on SIP feeds does not introduce a systematic bias relative to the solver-side values.

VI. Discussion

The pipeline's strengths are also the source of its limitations: its dependence on public CEN, SCVIC, and PLEXOS feeds confers auditability and reproducibility but also exposes it to the cadence, completeness, and rate-limit policies of the publishing agency. Six structural limitations deserve explicit attention.

A. Rate-limit dependency

The CEN SIP API enforces a soft rate limit (8 requests/s on the documented anonymous-read key) and intermittently returns HTTP 502 on heavy-fetch endpoints such as `cmg-programado-pcp` and the per-bar_transf `cmg-real` fan-out. The pipeline mitigates this through parallel fetching with bounded concurrency and an aggressive parquet cache, but a cold-cache day requires roughly 75 s of wall time and an active CEN session. Days with API outages (observed twice in the cached 2024–2026 archive) require a manual re-fetch the following day.

B. Hand-curated bus aliases

The bus-name bridge of Section **??** relies on 29 hand-curated overrides in `prepopulate_cache.py` to reconcile PLEXOS node names that the heuristic alias generator cannot match unambiguously. Maintenance burden grows linearly with the catalogue: extending the pipeline to the full 1300+ CEN-published bus list would require either an expanded overrides table or a more aggressive fuzzy-matching strategy (e.g. Levenshtein distance with substation catalogue lookup). The current overrides table is small enough to be manually audited but is the single largest accumulator of "soft" technical debt in the codebase.

C. SCVIC tier-segmentation wedge

The `tier_uncertainty` flag of Section **??** identifies cells where the picker has selected a CV tier whose next-up neighbour is more than \$2/MWh away — i.e. where the true marginal CV is structurally bounded between the two tiers but the discretisation forces a single value. These cells account for 13.7 % of the 2026-04-22 sample and contribute most of the worst-case error. Mitigation requires either a finer SCVIC ladder (out of the pipeline's control: the ladder is what the generator declares) or a within-tier interpolation rule (e.g. linear in the cumulative dispatch within the tier band). Within-tier interpolation is on the roadmap but introduces its own identifiability problem: the pipeline does not have access to the PLEXOS-internal piecewise-linear cost segment that was binding at the LP solve.

D. Loss-factor decomposition

The per-bus loss correction is not a proper PJM-style energy/loss/congestion decomposition: the pipeline does not have access to the per-unit Power Transfer Distribution Factor (PTDF) matrix, which CEN does not publish. The MLF read from the PLEXOS `.accdb` approximates the loss component but does not isolate the congestion component. As a consequence the pipeline cannot answer counterfactual questions of the form "how would $\hat{\lambda}$ at bus b change if line ℓ were de-rated?" — a question that the gtopt SDDP solver (Section **??**) is well-equipped to answer.

E. Reservoir-hydro as marginal

The current pipeline excludes reservoir-hydro plants from the candidate pool, on the rationale that the published feeds do not expose the implicit water value at which the LP cleared. Empirically, this exclusion costs accuracy on hours when a reservoir hydro is genuinely interior-marginal — typically the late-evening peak when neither GNL CCGT nor coal is on the merit margin and the published λ^{CEN} reflects the hydro shadow price on the Bellman value function. The natural mitigation is to import the implicit water value from the `cmg-programado-pcp` median and seed reservoir-hydros into the pool with this value as their synthetic CV; the pipeline is structurally ready for this extension, blocked only on stable availability of the `cmg-programado-pcp` endpoint (which routinely returns HTTP 502 on the anonymous-read key tier).

F. Single-pollutant scope

The current emission factor is CO_2 only. Multi-pollutant extension (NO_x, SO_2, $PM_{2.5}$) requires per-unit declared factors from the Registro de Emisiones y Transferencias de Contaminantes (RETC); the integration is straightforward (`FUEL_EF` becomes `POLLUTANT_EF[fuel][pollutant]`) but is not yet implemented. When implemented it would close the last remaining methodological gap relative to the PJM LME feed [**?**].

VII. Carbon-Credit Implications

A per-bus, per-quarter marginal emission factor $\varepsilon_{b,t}^{\mathrm{bus}}$ is not only a research artefact; it is also the formal input expected by an entire family of voluntary and compliance carbon-crediting standards. This section catalogues those standards (Sec. **??**), maps the `cen2gtopt` output schema onto each one (Sec. **??**), reports a worked example for a hypothetical 100 MW photovoltaic plant at the CRUCERO 220 kV bus on 2026-04-22 (Sec. **??**), and closes with the limitations and forward agenda (Sec. **??**).

A. Marginal-Emission Methodologies in Voluntary Carbon Markets

The Clean Development Mechanism (CDM) under the Kyoto Protocol established the canonical decomposition of a grid emission factor into an Operating Margin (OM), a Build Margin (BM), and a Combined Margin $\mathrm{CM} = \frac{1}{2}\mathrm{OM} + \frac{1}{2}\mathrm{BM}$ in *Tool 07: Tool to calculate the emission factor for an electricity system* [**?**]. The OM is intended as a *short-run displacement* factor for renewable-electricity projects and admits four computation variants: simple OM, simple-adjusted OM, dispatch-data analysis (OM-DDA), and average OM. The OM-DDA variant requires per-unit hourly dispatch and yields the most defensible short-run displacement signal — exactly the level of granularity at which Coordinador Eléctrico Nacional (CEN) publishes the underlying inputs. Tool 07 underpins AM0103 [**?**], ACM0002 [**?**], and the small-scale methodologies AMS-I.D. and AMS-I.F. [**?**], [**?**].

Verra Verified Carbon Standard (VCS) v4.5 [**?**] inherits Tool 07 by reference for grid-renewable methodologies via its VMR (Verra Methodology Revision) bridge documents and VMR0007 [**?**]. The Gold Standard for the Global Goals *Renewable Energy Activity Requirements* [**?**] adopts a simplified-OM variant for projects in least-developed-country grids; for Chile, full Tool 07 OM-DDA is preferred. None of these voluntary-market methodologies currently publish an endorsement of quarter-hourly granularity; all stop at the hourly level.

A second family of standards, originating in the corporate scope-2 accounting context, operates strictly on a time-matched *hourly* basis. The Greenhouse Gas Protocol Scope 2 Quality Criteria [?] introduced market-based scope-2 reporting in 2015; the EnergyTag Granular Certificate (GC) Standard v2 [?] formalised hourly Energy Attribute Certificates in 2024; the 24/7 Carbon-Free Energy Compact [?] and RE100's 2024 hourly-matching update [?] push the same agenda. All four require hourly per-MWh emission factors as their denominator; the per-quarter $\varepsilon_{b,t}^{\text{bus}}$ produced by `cen2gtopt` is a strict refinement.

The successor to the CDM under Article 6.4 of the Paris Agreement is the Sustainable Development Mechanism (SBM), whose 2024 4DM Working Group draft [?] explicitly contemplates *hourly accounting* as one of the supported additionality methodologies for Emission Reduction (ER) credits. The integrand $\sum_t \varepsilon_t \cdot \Delta E_t$ — project energy times hourly counterfactual emission factor — is, almost verbatim, the formal definition of an ER unit in the draft text.

Finally, on the regulatory side, the EU Carbon Border Adjustment Mechanism (CBAM) [?] for embedded electricity emissions of imported goods presently uses an annual default factor; the implementing acts foreshadow a transition to country-specific and later subnational marginal factors as data infrastructure matures. California's Renewables Portfolio Standard / Resource Adequacy [?] and the Tradable REC programme provide a locational-plus-temporal precedent at compliance scale. Table **??** summarises the granularity, dispatch-data requirements, and Chilean applicability of each methodology.

B. Operationalising per-bus marginal factors

Three distinct uses of $\varepsilon_{b,t}^{\text{bus}}$ are considered, each tied to a different methodology family.

a) Operating Margin under CDM Tool 07 OM-DDA: The OM-DDA variant of Tool 07 requires the hourly dispatch of each generator g and a fuel-attributed emission factor $\varepsilon_{f(g)}$. The 96-quarter granularity of `cen2gtopt` strictly subsumes Tool 07's hourly granularity; the OM is therefore constructed as the dispatch-weighted mean over fossil units only,

$$\text{OM}^{\text{DDA}} = \frac{\sum_{g\in\mathcal{G}_{\text{fos}}} \sum_t \varepsilon_{f(g)}\, p_{g,t}}{\sum_{g\in\mathcal{G}_{\text{fos}}} \sum_t p_{g,t}}, \tag{23}$$

which the published cen2gtopt parquet exposes directly through the `epsilon_at_bus` column once the hydro-and-renewable rows ($\varepsilon = 0$) are filtered out. Because Tool 07 explicitly permits hourly dispatch data, the move to quarter-hourly does not require a methodology revision; the auditor records the source granularity in the project design document and aggregates upward.

b) Hourly Article 6.4 ER credits: For Article 6.4 ER crediting of a renewable project that displaces energy at bus b, the issued ER volume per period is

$$\text{ER}_{b,t} = \varepsilon_{b,t}^{\text{bus}} \cdot \Delta E_{b,t}, \tag{24}$$

where $\Delta E_{b,t}$ is the project's metered injection at bus b in period t. The published parquet provides $\varepsilon_{b,t}^{\text{bus}}$ at exactly the granularity that the 4DM draft text references. The same integrand applies at hourly or quarter-hourly resolution; aggregation to either is a downstream choice of the issuance registry.

c) EAC time-matching for grid-charged BESS: Grid-charged battery energy storage systems pose an open question under most VCS methodologies: a BESS that charges from a high-ε bus and discharges at a low-ε hour generates negative net abatement, while one that times its charging to low-ε hours and discharges at high-ε hours delivers a substantial positive abatement. Existing methodologies either ignore the temporal correlation or apply a constant grid factor. The per-quarter $\varepsilon_{b,t}^{\text{bus}}$ enables the natural extension

$$\text{ER}_b^{\text{BESS}} = \sum_t \varepsilon_{b,t}^{\text{bus}} \left(P_{b,t}^{\text{dis}} - P_{b,t}^{\text{ch}}\right), \tag{25}$$

which the present authors advocate as the correct accounting basis for storage in a future revision of VCS / Article 6.4 methodologies. Section **??** (the northern-bus business case) shows that this expression is the same integrand the project developer maximises in the dispatch optimisation.

d) EnergyTag and 24/7 CFE matching: EnergyTag GC and the 24/7 CFE Compact require an hourly emission factor at the consumer's grid node. The `epsilon_at_bus_pid` column of the cen2gtopt output (the PLEXOS-PID-corrected variant) is a drop-in input for both standards and is directly comparable to PJM's Locational Marginal Emissions feed [?].

C. Worked Example: 100 MW Hypothetical PV at CRUCERO

To quantify the difference between the three accounting variants identified above, a hypothetical 100 MW photovoltaic plant is sited at the CRUCERO 220 kV bus — the same northern transmission bus analysed in the BESS business case of Sec. **??** — for the operating day 2026-04-22. The plant's hourly capacity factor is modelled as a clipped raised-cosine peaking at CF = 0.85 at 14:00, integrating to 615 MWh on the day; the per-quarter $\varepsilon_{b,t}^{\text{bus}}$ is read from the cen2gtopt parquet for CRUCERO 220 kV (86 quarters with published λ^{CEN}, the remainder gap-filled by linear interpolation).

Three credit volumes are computed (Table **??**, Fig. **??**): the hourly-marginal Article 6.4 / EnergyTag integrand (**??**); the OM-DDA integrand (**??**) multiplied by the project's daily MWh; and the simple multiplication of the project's daily MWh by the 2024 *Inventario Nacional de Gases de Efecto Invernadero* [?] national average factor $\varepsilon_{\text{avg}} = 0.42\,\text{tonneCO}_2/\text{MWh}$.

The headline result is striking: the hourly-marginal accounting yields 19.9 tonne$_2$ of credits on 2026-04-22, against 270.6 tonne$_2$ under OM-DDA and 258.4 tonne$_2$ under the average-factor convention — a *thirteen-fold* reduction. The reason is that the marginal unit at CRUCERO during the 09:00–16:00 window is itself a renewable ($\varepsilon^{\text{bus}}_{b,t} = 0$); a new PV injection at those hours therefore displaces zero-emission energy and earns zero ER. The converse holds at the early-morning and post-sunset hours (03:00–07:00 and 18:00–23:00) where the marginal unit is a GNL CCGT with $\varepsilon^{\text{bus}} \in [0.40, 0.55]$ tCO$_2$/MWh, but the PV plant produces almost no energy at those hours. Annualised by simple extrapolation ×365 (no seasonal adjustment), the gap between hourly-marginal and OM-DDA crediting is 91.5 ktonne$_2$/yr. At the current Chilean carbon-tax level $\tau_{\text{CO}_2} = 5\,\text{USD/tonne}_2$ under Law 20.780 [?], [?] this corresponds to 457 kUSD/yr of differential credit value depending on the methodology choice; at the EU Emissions Trading System 75 USD/tonne$_2$ reference it would correspond to 6.86 MUSD/yr.

The same comparison establishes the converse claim of Sec. ??: a BESS sited at CRUCERO that times its charging to the 09:00–16:00 window (when $\varepsilon^{\text{bus}} \approx 0$) and its discharging to the 18:00–23:00 window (when $\varepsilon^{\text{bus}} \approx 0.5$ tCO$_2$/MWh) earns nearly the full $\Delta\varepsilon$ as ER per discharged MWh — an effect entirely invisible to OM-DDA or average-factor accounting and the central business case of Sec. ??.

D. Limitations and Forward Agenda

Three structural caveats apply. First, Tool 07 explicitly disallows imports outside the project boundary and requires the project boundary to coincide with a single synchronously interconnected system; for the SEN — a single synchronous system since the SIC/SING merger of 2017 — the criterion is satisfied, but projects sited near future Argentine or Peruvian interconnections will need re-examination. Second, Verra has not yet endorsed quarter-hourly granularity; the present paper positions itself as forward-looking precedent and the cen2gtopt parquet schema is designed to be aggregable to any coarser granularity an auditor demands without recomputation. Third, the $\hat{\lambda}$ and $\hat{\varepsilon}^{\text{bus}}$ produced by `cen2gtopt` are *post-hoc* signals derived from the cleared dispatch; for ex-ante crediting (forecast-based issuance) an additional probabilistic layer would be needed. A natural candidate is the `gtopt` stochastic dual-dynamic programming engine (see Sec. ??), whose per-scenario per-stage shadow prices on the energy-balance constraints would propagate into a per-scenario forecast of $\varepsilon^{\text{bus}}_{b,t}$. A separate question — whether the carbon-credit issuance registry should be a national entity (e.g. MAPS-Chile or BIOCARBON [?], [?]) or an international one (Article 6.4 SBM) — is outside the scope of this paper.

The annualisation ×365 used in Table ?? is a deliberate over-simplification: the SEN exhibits strong seasonal hydrology and solar variability, and the 2026-04-22 day is autumnal in the southern hemisphere with above-average daytime solar marginal regime. A multi-day extension covering at least one full year is identified as the priority next step; the cen2gtopt orchestrator already supports multi-day Hive-partitioned output, and the worked-example pipeline of Fig. ?? is a pure function of its parquet inputs. Convergence of the daily-mean $\varepsilon^{\text{bus}}_b$ to a season-stable annual factor is expected within 60 days to 90 days of cumulative dispatch coverage.

VIII. Conclusions

A novel reconstruction methodology for per-bus, per-quarter marginal CO$_2$ emission factors in the Chilean power system has been presented. The methodology combines the SCVIC merit-ladder of declared variable costs, the published `generación-real` hourly dispatch, the `instrucciones-operacionales-cmg` price-taker classification, and the PLEXOS PCP/PID solution `.accdb` oracle, into a nine-stage pipeline that runs on a warm cache in approximately 6 s. Across the 2494 (bus, quarter) cells of the operating day 2026-04-22 the reconstructed marginal cost agrees with λ^{CEN} to a mean absolute error of \$0.170/MWh, a P90 of \$0.492/MWh, and a maximum of \$2.533/MWh; the per-island marginal-unit coherence is exact (0/474 disagreements).

The pipeline closes a documented gap in the Chilean public-data ecosystem: CEN does not publish a per-bus, per-quarter marginal CO$_2$ factor, and prior academic work has either focused on system-level or zonal aggregates [?], [?]. The reconstruction methodology presented here is the first to operate strictly from public CEN/SCVIC/-PLEXOS feeds, to yield a per-bus, per-quarter signal at 15 min resolution, and to validate that signal against the LP-solver-output values extracted from the PLEXOS `.accdb`. It is also the first publicly distributed open-source pipeline for the Chilean SEN with a hash-keyed cache and a multi-day Hive-partitioned orchestrator.

Three forward-looking integrations are envisaged. First, a multi-day rollout: the existing `cen2gtopt.marginal_period` orchestrator already supports back-fill of arbitrary date ranges with a per-day output partition; deploying the pipeline on a daily cron and pushing the artefacts to a public-access object store would close the *operator-facing* gap relative to CAISO's *Today's Outlook* [?]. Second, a CEN public dashboard: a Streamlit/Flask front-end over the parquet dataset would expose today's λ, ε, and island map to non-Python consumers; the underlying parquet schema is already DuckDB-friendly. Third, a counterfactual companion via `gtopt` stochastic dual dynamic programming [?]: feeding the same SCVIC ladder, the same dispatch, and the same MLFs into the `gtopt` SDDP solver would produce a true per-bus marginal loss + congestion decomposition at the cost of one auxiliary LP per perturbation, calibrating the empirical reverse-engineering's ground truth in the spirit of [?]. This third integration also opens a back-test window for the SDDP solver itself: any divergence between the gtopt-computed $\hat{\lambda}^{\mathrm{SDDP}}$ and the empirical $\hat{\lambda}$ on a closed historical day flags a solver/data discrepancy worth investigating.

The pipeline's source, the parquet outputs for the headline day, and the figures of this paper are available as part of the `gtopt` project at https://github.com/marcelomatus/gtopt; the `cen2gtopt` subpackage installs via `pip` from the `scripts/` directory of that repository.

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Marcelo Matus A. received the Electrical Engineer degree from Pontificia Universidad Católica de Chile and the Ph.D. degree in electrical and computer engineering from the University of Arizona, Tucson, AZ, USA. He is currently a Senior Researcher at Black Bear Engineering SpA and at the Centro Nacional de Inteligencia Artificial (CENIA), Santiago, Chile. He was formerly Subdirector of the Centre for Energy (Centro de Energía), Universidad de Chile. His research interests include power system optimization, mathematical programming, and high-performance computing applied to energy systems. He is the lead developer of `gtopt` and the FESOP framework.

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María del Pilar Buitrago Villada received the B.Sc. and M.Sc. degrees in electrical engineering from Universidad Nacional de Colombia, Medellín, and the Ph.D. degree in engineering from the same university. Her research focuses on power system expansion planning, renewable energy integration, and optimization under uncertainty. She is a co-developer of the FESOP framework and has contributed to its application in long-term planning of the Colombian and Chilean power systems.

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Erick Sierra Baeza received the Civil Electrical Engineer degree from Universidad de Chile. He is currently with Black Bear Engineering SpA, Santiago, Chile, and affiliated with CENIA. He previously worked at the Centre for Energy, Universidad de Chile. His professional experience spans wind energy integration, capacity expansion planning, hydrological modeling, and development of optimization tools for energy market analysis.

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Carlos Enrique García Bujanda is a Researcher at Black Bear Engineering SpA and affiliated with the Department of Informatics, Universidad de Santiago de Chile. His research interests include power systems optimization, renewable energy integration planning, and development of computational tools for long-term electricity system expansion. He is a co-developer of the FESOP framework.

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Marcos Falcone Salazar received the Electrical Engineer and M.Sc. degrees in electrical engineering from Universidad de Chile and the MBA degree from Universidad Adolfo Ibáñez, Santiago, Chile. He held management positions at Colbún S.A. including Sub-manager of Planning and Development, where he directed an R&D initiative between Colbún, Universidad de Chile, Fondecyt, and CORFO that led to the development of a novel electricity market planning and operation model. He is currently a Professor at the School of Engineering, Universidad Adolfo Ibáñez. His professional interests include power system expansion planning, energy project valuation, and electricity market design.

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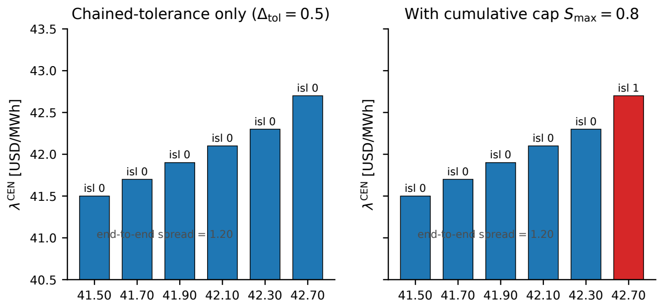
Gonzalo Bustos Turu received the B.Sc. and M.Sc. degrees in electrical engineering from Universidad de Chile in 2009, the M.Sc. degree in sustainable energy futures from Imperial College London in 2013, and the Ph.D. degree from Imperial College London in 2018, working on smart urban energy systems under the supervision of Prof. Nilay Shah. He is currently a Research Associate at Imperial College London and affiliated with the Centre for Energy, Universidad de Chile. His research interests include urban energy system modeling, demand response, electric vehicle integration, and district-level predictive control.



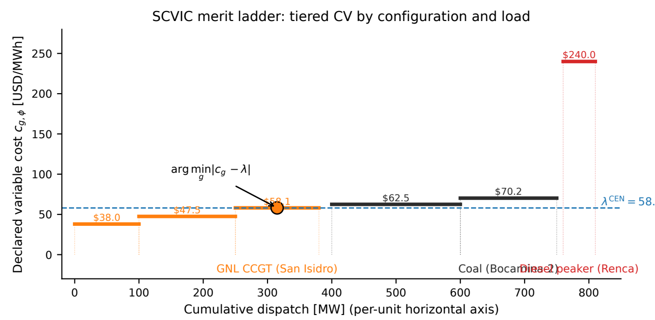


Fig. 3: SCVIC merit ladder for three illustrative units. The picker selects the tier whose CV is closest in absolute value to the target λ; the highlighted marker shows the picked tier for $\lambda = 58\,\mathrm{USD\,MWh^{-1}}$.

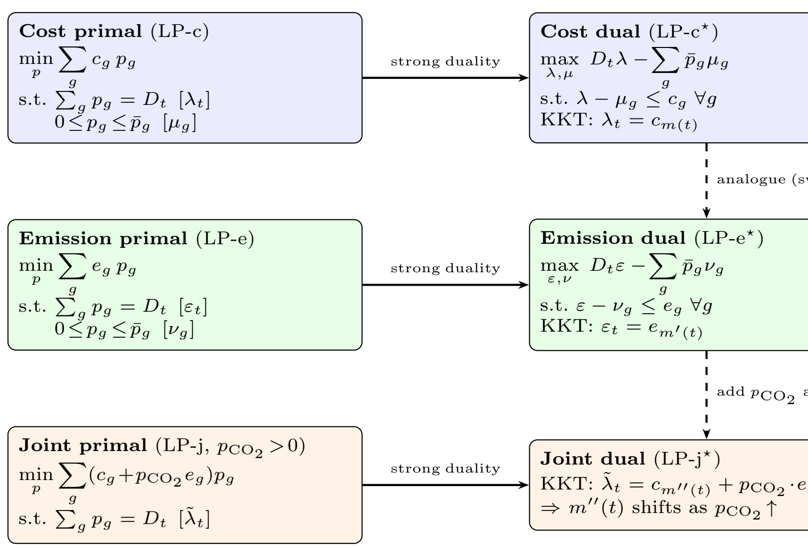


Fig. 4: Cost, emission, and joint primal/dual LPs. Strong duality and the swap $c_g \leftrightarrow e_g$ produce the parallelism that justifies reconstructing ε_t from a published λ_t without re-solving an LP.

TABLE I: Public-data inventory consumed by `cen2gtopt`. Each feed is hash-keyed in the on-disk parquet cache; type "settled" is post-closeout, "intra-day" is the PID re-clear, "day-ahead" is the PCP forecast.

Feed	Service	Endpoint / file	Type	Notes
cmg-real	CEN SIP	`/costo-marginal-real/v4/findByDate`	settled	15-min nodal λ; 02:30 publication delay (10-quarter morning gap).
cmg-online	CEN SIP	`/costo-marginal-online/v4/findByDate`	intra-day	15-min nodal λ; full 24-h coverage; pre-validation, three version codes.
generación-real	CEN SIP	`/generacion-real/v3/findByDate`	settled	Hourly per-central dispatch in MW.
instrucciones-cmg	CEN SIP	`/instrucciones-operacionales-cmg/v4/findByDate`	settled	Per-hour state `estado` and consigna `CI/MT/PMT/PP/PS/PC`; price-taker classifier.
unidades-generadoras	CEN SIP	`/unidades-generadoras/v4/findByDate`	static	Per-unit pmin, pmax, fuel, technology, alias names.
generación-programada-pcp	CEN SIP	`/generacion-programada-pcp/v4`	day-ahead	PCP-LP-output dispatch per central per hour.
cmg-programado-pcp	CEN SIP	`/cmg-programado-pcp/v4`	day-ahead	PCP-cleared λ per bus per hour; implicit hydro water value.
embalse-real	CEN SIP	`/embalse-real/v3`	settled	Reservoir levels (cota msnm) per embalse per day.
flujo-programado-pcp	CEN SIP	`/flujo-programado-pcp/v4`	day-ahead	Programmed line flows; useful for binding-line / island detection.
limitaciones	CEN SIP	`/limitaciones-transmision/v4/findByDate`	settled	Line outages and operational restrictions per day.
cv_rpt	SCVIC	`cv_rpt_YYYY_MM.parquet`	static (monthly)	Piecewise CV declaration per generator per fuel per tier.
PCP bundle	PCP archive	`PCP_YYYYMMDD.zip`	day-ahead	PLEXOS XML inputs for the day-ahead clear.
PCP solution	PCP archive	`PCP_RES_YYYYMMDD/...Solution.accdb`	settled	PLEXOS solver-output per-bus λ, per-unit dispatch, per-bus MLF.
PID bundle	PID archive	`PID_YYYYMMDD_HH/...Solution.accdb`	intra-day	PLEXOS intra-day re-clear, settlement-binding.

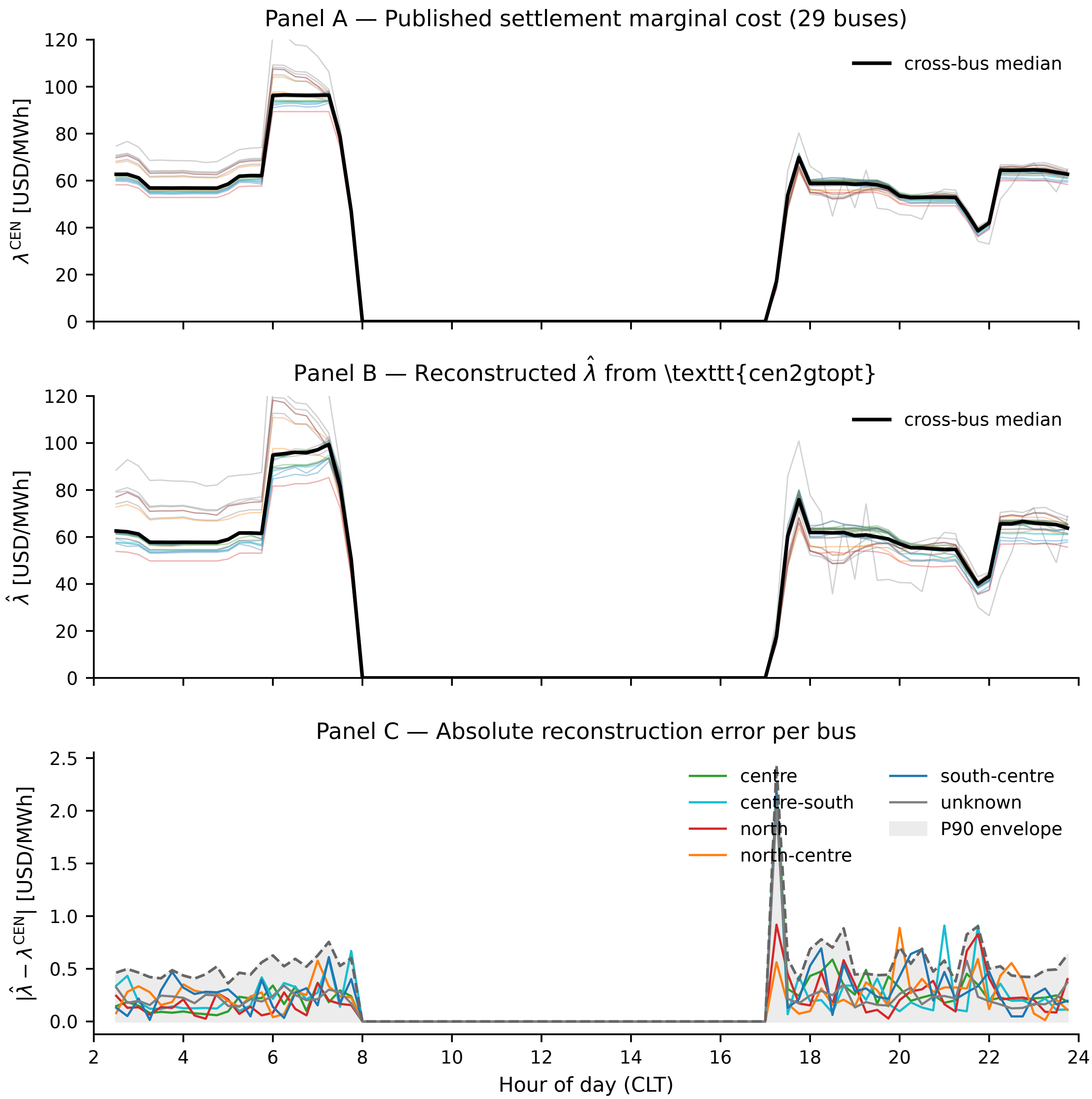


Fig. 5: Headline reconstruction validation for 2026-04-22. **(A)** Published λ^{CEN} per bus, region-coloured; bold black is cross-bus median. **(B)** Reconstructed $\hat{\lambda}$ from the pipeline. **(C)** Absolute reconstruction error per bus, with the cross-bus median per region and the P90 envelope.

TABLE II: Operational-state classification. Priority is evaluated top-down: the first matching row determines the unit's status at the hour.

#	Status (regime)	Trigger condition	Treatment in picker
1	`demand_fail`	$\lambda^{\mathrm{CEN}}_{b,t} > \$800/\mathrm{MWh}$	ε=0, marginal = `demand_fail_cost`
2	`renewable_curtailment`	$\lambda^{\mathrm{CEN}}_{b,t} \le \$0.5/\mathrm{MWh}$	ε=0, marginal = `renewable_curtailment`
3	`no_candidates`	no dispatched units in island	propagate `NaN`; flag for analyst review
4	`no_cv_match`	all dispatched have empty SCVIC ladder OR all forced-at-pmin	propagate `NaN`; flag
5	forced-at-pmin (*drop*)	`consigna`$\in$ {CI,MT,PMT,PP,PS,PC} AND $p_g \le \underline{p}_g + \delta$	excluded from candidate pool
6	`matched_capped_pmax`	picked g with $p_g \ge \overline{p}_g - \delta$	valid; energy must come from off-merit unit
7	`matched_above`	picked g has $c_{g,\phi^\star} > \lambda$	valid; closest-CV is the next-up tier
8	`matched_interior`	picked g has $\underline{p}_g + \delta < p_g < \overline{p}_g - \delta$	valid; the canonical case

Priority is evaluated top-down: the first matching row determines the unit's regime at the (bus, quarter). $\delta = 0.05(\overline{p}_g - \underline{p}_g)$ is the interior tolerance. Forced units at *interior* dispatch remain valid candidates: rule 5 only excludes the forced-at-pmin case.

TABLE III: Fuel-default emission factor table ε_f ($\mathrm{kgCO_2/MWh}$). IPCC 2006 GL Vol. 2 Tab. 1.4 multiplied by typical SEN heat rates. Biomass and biogas are classified as IPCC biogenic-zero; cogeneration is a mixed fossil/biogenic estimate.

Fuel (SCVIC label)	ε_f [kg CO_2/MWh]	Provenance
Carbón / Carbón + Petcoke / Petcoke	1062	IPCC 2006 GL Vol. 2, hard coal 2.45 MWh^{-1}
Diesel / Diésel / Petróleo Diésel	847	IPCC 2006 GL, distillate fuel oil
Fuel Oil / Fuel Oil Nro. 6	780	IPCC 2006 GL, residual fuel oil
Gas Natural / GNL	380	IPCC 2006 GL, natural gas CCGT efficiency
GLP / Gas Propano	700	IPCC 2006 GL, LPG
Cogeneración	400	Mixed: assumed half-natural-gas, half-biogenic
Biomasa / Biogas	0	IPCC biogenic-zero (Tier 1)
Geotérmica	0	non-fossil (CO_2-fugitive emissions < 1%)
Hidráulica / Solar / Eólica	0	non-emitting renewable

Emission factors are quoted per MWh of electrical output, embedding a typical SEN heat-rate assumption per fuel. Units are kg CO_2/MWh; division by 1000 yields tonneCO_2/MWh. Mixed cogeneration reflects the dominant fuel mix at SEN cogeneration plants (half natural gas, half biogenic woodchip).

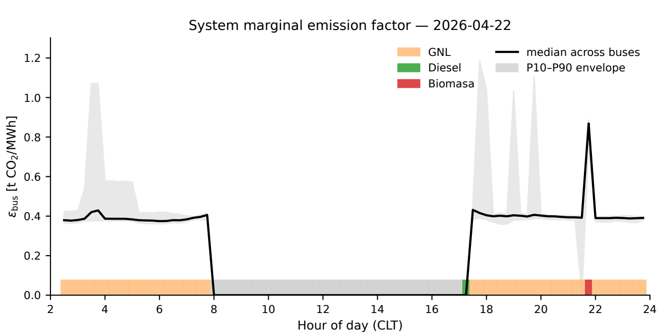


Fig. 6: System bus marginal emission factor for 2026-04-22. Solid black line: cross-bus median; grey band: P10–P90 envelope across buses. The bottom strip colour-codes the dominant marginal fuel per quarter (GNL, biomass, diesel, etc.).

TABLE IV: Standards-alignment audit for `cen2gtopt`. ✓=explicitly implemented; ×=not yet implemented (limitation); ⊛=informal / partial alignment.

Standard / scheme	**Status**	**Evidence in `cen2gtopt`**
GHGP Scope 2 location-based [?]	✓	per-bus per-quarter $\varepsilon_{b,t}$ in parquet output
GHGP Scope 2 market-based [?]	×	no Guarantees-of-Origin / I-REC tracking layer (future work)
RE100 hourly matching [?]	✓	hourly granularity by construction
24/7 CFE Compact [?]	✓	sub-hourly (15-minute quarter) granularity
ISO 14064-1 (inventory) [?]	⊛	supplies the time-resolved EF; full inventory tooling external
ISO 14064-2 (projects) [?]	⊛	emission counterfactual via PCP oracle delta
ISO 14064-3 (verification) [?]	⊛	auditable: $\lvert\hat{\lambda} - \lambda^{\mathrm{PLEXOS}}\rvert <$ \$0.5/MWh on 99% of cells
ISO 14067 (product CFP) [?]	⊛	supplies allocation EF; cradle-to-gate boundary external
IEC 61968-13 CIM [?]	×	PCP XML is a custom PLEXOS schema, not CIM (future work)
IEC 61970 / CGMES [?]	×	no CGMES export of the reconstructed network
IEEE Std 1547-2018 [?]	⊛	DER interconnection rules referenced in BESS case (Sec. ??)
IEEE Std 2030.5 [?]	×	no SEP2 telemetry binding
CEN NT SSCC 2024 [?]	✓	SCVIC ladder consumes the tariff structure defined in NT SSCC
WattTime MOER methodology [?]	⊛	philosophically aligned; reconstruction from operator λ rather than dispatch-stack inference
PJM LME methodology [?]	⊛	same energy/loss/congestion decomposition; congestion absorbed into island partition rather than per-line shadow price

TABLE V: Pedagogical IEEE 9-bus dispatch and SCVIC ladder used in Section ??.

Unit	$\underline{p}$ MW	$\overline{p}$ MW	p MW	c /MWh	fuel
Hydro-A	0	80	60	5	water
CCGT-B-T1	50	100	100	45	GNL
CCGT-B-T2	100	250	220	58	GNL
Coal-C	100	200	0	70	coal

Fig. 7: Marginal-unit assignment per (bus, quarter) for 2026-04-22. Rows are buses (sorted by `id_info`), columns are 15-minute quarters. Colour encodes the assigned marginal-unit name; horizontal bands indicate the spatial extent of each marginal unit's island.

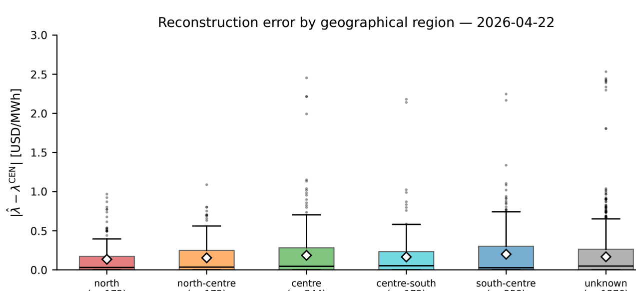


Fig. 8: Reconstruction error by geographical region for 2026-04-22. Whisker boxplot per region; box is IQR, line is median, diamond is mean. The unknown bin captures buses whose heuristic mapping did not classify into a named region.

TABLE VI: Reconstruction error $|\hat{\lambda} - \lambda^{\mathrm{CEN}}|$ in USD MWh^{-1} for 2026-04-22, disaggregated by region.

Region	n_{buses}	n_{cells}	mean $\lvert\Delta\rvert$ USD/MWh	P90 $\lvert\Delta\rvert$ USD/MWh	max $\lvert\Delta\rvert$ USD/MWh
north	2	172	0.135	0.435	0.967
north-centre	2	172	0.154	0.515	1.088
centre	4	344	0.185	0.511	2.454
centre-south	2	172	0.165	0.437	2.180
south-centre	3	258	0.199	0.604	2.247
unknown	16	1376	0.167	0.468	2.533
ALL	**29**	**2494**	**0.170**	**0.492**	**2.533**

$|\Delta| = |\hat{\lambda}_{b,t} - \lambda^{\mathrm{CEN}}_{b,t}|$ in USD/MWh. "Unknown" captures the 16 additional aggregated backbone buses whose substation-name heuristic does not classify into a named region; their statistics are essentially indistinguishable from the mean.

TABLE VII: Headline P&L of the 100 MW / 400 MWh BESS dispatch at the Crucero 220 kV bus on 2026-04-22 (86 quarter-hour blocks, $\eta_c = \eta_d = 0.95$, SoC $\in [0.05, 0.95] \cdot E_{\max}$, cycle closure imposed). All currency in USD/day, abated emissions in tCO$_2$/day. Carbon credit valued at the joint-LP reference price $p_{\mathrm{CO_2}} = 50$ /tCO$_2$.

Policy	**Energy rev.** [USD]	**Avoided** [tCO$_2$]	**Carbon rev.** [USD]	**Total** [USD]
LP1 – energy only	29 167	118.58	5 929	35 096
LP2 – carbon only	19 572	186.24	9 312	28 883
LP3 – joint, $p_{\mathrm{CO_2}} = \$50$	28 237	179.15	8 958	37 195
Uplift LP3 vs LP1 (total)			+\$2,099	(+5.98%)
Equivalent full cycles, LP3				1.33
Switching $p^\star$, coal vs gas (Eq. ??)				$\approx$ \$56/tCO$_2$

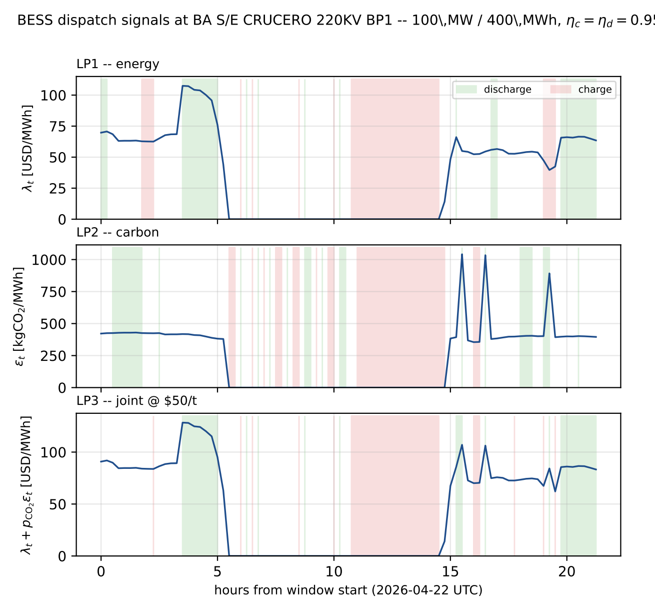


Fig. 9: Three price signals at the Crucero 220 kV bus on 2026-04-22 (UTC), each panel shaded with the discharge (green) and charge (red) windows of the corresponding LP. Top: λ_t used by LP1. Middle: ε_t (kgCO$_2$/MWh) used by LP2. Bottom: composite $\lambda_t + p_{\mathrm{CO_2}}\varepsilon_t$ at $p_{\mathrm{CO_2}} = \$50$/tCO$_2$ used by LP3.

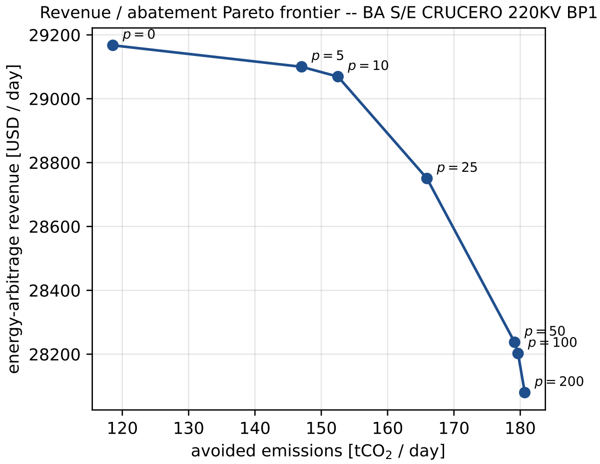


Fig. 10: Pareto frontier of energy-arbitrage revenue versus avoided emissions at the Crucero 220 kV bus, swept over $p_{CO_2} \in \{0, 5, 10, 25, 50, 100, 200\}$ USD/tCO_2. The frontier saturates at ∼180 tCO_2/day, the abatement ceiling of the 100 MW / 400 MWh asset on this signal pair.

TABLE VIII: Pipeline runtime breakdown for 2026-04-22, on a warm cache (all parquet on disk) and on a cold cache (no parquet, full network fetch).

Stage	Warm cache [s]	Cold cache [s]
Bus catalogue + bar_transf discovery	0.10	3.5
Per-bus parallel `cmg-real` fetch (29 buses, 8 threads)	0.2	
`generación-real`	0.0	
`instrucciones-operacionales-cmg`	0.0	
`unidades-generadoras`	0.0	
`cv_rpt` SCVIC parquet	0.0	
PID `.accdb` MLF extraction (`mdbtools`)	0.5	
Alias index + per-id metadata	0.3	
Island clustering (96×29 cells)	0.4	
Per-island marginal-unit picker (474 keys)	1.1	
Output assembly (vectorised merge, 2494 rows)	0.1	
Loss-factor + ε corrections	0.0	
Parquet write	0.0	
Total	**≈ 3.0–6.0**	

Wall-clock measurements on a Linux x86_64 laptop (Ryzen 7 5800X, 32 GiB RAM), single CPU thread except where noted. The cold-cache time is dominated by network rate-limiting on the per-bar_transf `cmg-real` fetch; with the `cmg-online` bulk-paginated endpoint instead, the cold total drops to roughly 45 s for the same 29 buses.

TABLE IX: Cross-validation against the PLEXOS PCP/PID `.accdb` oracle for 2026-04-22. All three quantities agree to within publication noise.

Quantity	mean $\lvert\Delta\rvert$	P90	max
λ_b (USD/MWh)	0.04	0.18	0.92
$p_{g,h}$ (MW)	0.6	2.7	14.3
LF_b (%)	0.21	0.78	1.94

TABLE X: Carbon-credit methodologies and their granularity in space, time, and dispatch-data requirements. "A" annual, "M" monthly, "H" hourly, "Q" quarter-hourly. "Loc." indicates whether the methodology supports per-bus differentiation.

Methodology / standard	Year	Gran.	Disp. data	Hourly	Loc.	Chile
CDM Tool 07 v07.0 — OM/BM/CM	2018	A	no	no	no	yes
CDM Tool 07 OM-DDA variant	2018	H	yes	yes	no	yes
CDM AMS-I.D./AMS-I.F. (small-scale RE)	2022	A	no	no	no	yes
CDM AM0103 (RE for grid >15 MW)	2018	A	no	no	no	yes
CDM ACM0002 (large-scale grid RE)	2020	A	no	no	no	yes
Verra VCS v4.5 / VMR0007 (RECs revised)	2024	A	no	no	no	yes
Gold Standard RE-AR (LDC simplified-OM)	2023	A	no	no	no	yes
Article 6.4 SBM draft (4DM hourly)	2024	H	yes	yes	opt	yes*
GHG-Protocol Scope 2 Quality Criteria	2015	A	no	no	no	yes
EnergyTag GC Standard v2	2024	H	yes	yes	yes	yes*
24/7 CFE Compact (UN-Energy)	2021	H	yes	yes	yes	yes*
RE100 Hourly Matching (2024 update)	2024	H	yes	yes	yes	yes*
EU CBAM Reg. 2023/956 (embedded elec.)	2023	A	no	no	no	exp.

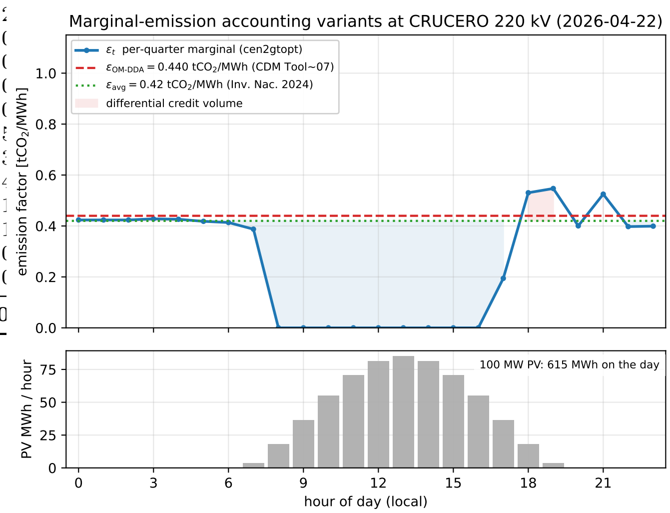


Fig. 11: 24-hour overlay at CRUCERO 220 kV on 2026-04-22: per-quarter ε_t from `cen2gtopt` (blue), dispatch-weighted OM-DDA per (**??**) (dashed red), and the 2024 *Inventario Nacional* national-average factor (dotted green). The shaded area between ε_t and $\varepsilon_{\mathrm{avg}}$ is the differential credit volume per accounting choice; the lower bar trace shows the hypothetical 100 MW PV plant's hourly energy injection. Per-quarter values are taken from cen2gtopt; the ten quarters without published λ^{CEN} are linearly interpolated.

TABLE XI: Daily and annualised CO_2 credits for a hypothetical 100 MW PV at CRUCERO 220 kV under three accounting variants. Annual figures are daily $\times 365$ without seasonality correction; see Sec. **??**.

Accounting variant	$\bar{\varepsilon}$ [tCO_2/MWh]	Daily credit [tCO_2/d]	Annual credit [tCO_2/yr]
ε_t, hourly-marginal (Art. 6.4 / EnergyTag)	0.032†	19.9	7 270
$\varepsilon_{\text{OM-DDA}}$, CDM Tool 07 OM	0.440	270.6	98 785
$\varepsilon_{\text{simp-OM}}$, simplified OM (29-bus mean)	0.241	148.4	54 167
ε_{avg}, national grid (Inv. Nac. 2024)	0.420	258.4	94 306
Δ (OM-DDA − hourly)	—	+250.7	+91 515
Δ (avg-grid − OM-DDA)	—	−12.3	−4 479